

Seismic behavior of self-centering precast segmental bridge piers with external auxetic steel shear panel dampers

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ABSTRACT

Precast self-centering segmental bridge piers have gained significant attention in bridge construction. Among them, development of the energy dissipation devices that are applicable for the self-centering bridge piers has been a major focus over the past three decades. This study develops a novel post-tensioned precast segmental bridge piers (SC-PSBPs), incorporating replaceable external auxetic steel shear panel dampers (ASSPDs) at the bottom and connecting steel plates at the intermediate segments. The seismic behavior of the SC-PSBPs is investigated through experiments and numerical simulations. Auxetic metamaterials are utilized to develop two kinds of perforated steel plates, including elliptical and peanut-shaped perforations, for the ASSPDs. Eight SC-PSBP specimens are tested to study failure patterns, hysteresis response, residual displacement, segment gap opening behavior, and local response of steel components. Test results show that the proposed SC-PSBP, equipped with unbonded post-tensioning (PT) tendons and ASSPDs, exhibits stable hysteresis response and self-centering capacity with minimal concrete damage. Plastic deformation is concentrated in the ASSPDs, while the intermediate connecting steel plates behave elastically throughout the loading history. A numerical model is developed to investigate the influence of design parameters on the hysteresis behavior of SC-PSBPs. Parametric analyses using the validated FE models explore the seismic behavior of SC-PSBPs with different geometries of ASSPDs, connecting steel plates, and initial PT forces. Numerical results show that both the ASSPDs and the connecting steel plates are imperative to achieve the desired rocking and damage concentration mechanisms of SC-PSBPs. The numerical findings also indicate that decreasing the porosity of ASSPDs effectively enhances the bearing and energy dissipation capacity of SC-PSBPs, as well as increases their residual displacement.

1. Introduction

Bridges play a crucial role in transportation infrastructure, and any service disruption can result in significant socioeconomic impacts [1,2]. As bridges may suffer severe damage from natural and man-made hazards (e.g. earthquakes, blast) during their service lives, there is a growing need for repair and replacement. Traditional cast-in-place reinforced concrete bridge columns often develop plastic hinges during seismic events, leading to permanent deformations and severe damage. They are thus unfavorable for post-earthquake repair and recovery of the bridge systems that require large recovery time and cost. In response, seismic resilient bridge structures have gained attention as an alternative to cast-in-place construction, which aims to provide the bridges with the capacity that operates normally without repair or can be restored after reduced repair after strong earthquake shaking [3, 4]. Among others, the self-centering precast piers offer advantages such as reduction of construction time and environmental pollution,

and improvement of maintenance efficiency. More importantly, they incorporate a rocking mechanism by unbonded post-tensioning (PT) tendons to reduce residual deformations and damage to the pier after earthquakes, which facilitate a rapid recovery of bridge operation [5].

Extensive studies have been reported on the seismic behavior of self-centering precast bridge piers [6–8]. For example, Palermo et al. [9] proposed a novel design of self-centering bridge piers with internal and external dissipation configurations. The quasi-static tests on the novel piers found that these columns suffered minimal damage and had negligible residual displacement when compared to monolithic counterparts. Similarly, Yamashita and Sanders [10] performed shake table tests on precast unbonded PT concrete columns. The results indicated minor residual displacements and limited concrete spalling at the base upon unloading. The dynamic behaviors [11,12] and higher mode demand [13] of the posttensioned precast segmental piers were other topics of great interest in the research area of self-centering

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precast bridge piers. However, the use of self-centering precast piers is mostly limited to low-seismicity areas as the seismic performance during strong earthquakes is not fully understood [14,15]. Although studies have shown that self-centering precast piers exhibit flag-shaped hysteresis loops, indicating good self-centering ability [16–18], they often lack sufficient energy dissipation capacity, which hinders their application in regions with moderate to high seismicity [19]. Furthermore, localized damage can still occur at the interface of intermediate segments in self-centering precast piers with low energy dissipation capacity.

In order to overcome these challenges in segmental bridge piers, researchers have explored methods to enhance the energy dissipation by using additional internal or external energy dissipation (ED) devices. A typical internal device attributes to the popular mild steel rods that are usually embedded in the foundation and protruding into the segments through the lower connections [20–23]. However, replacing internal ED rods can be difficult if they are damaged by a strong earthquake [24]. External devices, such as external mild steels [19], friction EDs [25], viscous damping devices [26], are commonly used in the joint region. For example, Chou and Chen [27] proposed a novel diagonally placed energy-dissipating device as external dissipators, which were shown to contribute to an increase in the equivalent damping ratio of PT composite bridge piers. Cai et al. [28] improved the energy dissipation capacity of a structure by installing steel top and seat angles near the beam–column joint in a self-centering precast concrete frame. While these external and internal ED devices have significantly enhanced the energy dissipation capacity, the self-centering segmental bridge piers still face challenges for practical application due to their complex configurations at the base of the piers. Damage to the pier still appeared at the base and the connection of the precast segments using the ED devices proposed by past studies [5,6,29]. Therefore, there is an urgent need to develop new ED devices that provide substantial energy dissipation capacity for self-centering precast piers while having simple configurations for easy construction and post-earthquake repairs.

Mechanical metamaterials are a class of man-made materials with extraordinary properties that can be found in naturally available materials [30]. One type of mechanical metamaterial that has received extensive study is negative Poisson's ratio materials, also known as "auxetics" [31–33]. Auxetics exhibit desirable mechanical properties, such as high energy absorption capacity [34,35], fracture resistance [36], and shear resistance [37]. Chen et al. [38] developed an innovative design for an oval perforated plate structure exhibiting auxetic effect and found that the novel structures with smaller Poisson's ratio had higher effective specific energy absorption. Zhang et al. [39] proposed a novel auxetic perforated plate used as the core of the buckling-restrained brace to explore the effect of auxetic structures on the seismic performance. Instead of relying on beam-dominated auxetic microstructures, researchers have discovered that incorporating an orthogonal elliptical hole pattern into solid materials can be advantageous in achieving an adjustable auxetic response. However, it should be noted that the presence of sharp holes often results in high stress concentration. To address this issue and inspired by the research reported in the literature [33,40], Zhu et al. [41] and Wang et al. [42,43] introduced the holes with smooth boundaries into the auxetic structures. Zhu et al. [41] incorporated a design of core plate with a square array of mutually orthogonal peanut-shaped holes into the buckling-restrained braces (BRBs), and Wang et al. [42,43] implemented a similar design into steel plate shear walls (SPSWs). These studies have shown that such perforated auxetic structures exhibit stable hysteresis performance and significant energy dissipation, indicating great potential of the perforated auxetic structures to be employed as the ED devices in SC-PSBPs. Following the studies by Zhu et al. [41] and Wang et al. [42,43], this paper employs the elliptical and peanut-shaped perforations into the design of auxetic ED devices. The purpose of this study is to (1) innovate an auxetic type of ED devices that can be easily installed, repaired, and replaced at the SC-PSBPs, (2) mitigate the damage to the

pier base and to the joint region between two intermediate segments, and (3) explore the effects of design parameters of the novel ED devices and the SC-PSBP on the seismic response of the novel pier components.

Motivated by the issues of SC-PSBPs mentioned above and the findings of prior research, this paper focuses on developing a new type of replaceable energy-dissipating device, namely auxetic steel shear panel dampers (ASSPDs), to address the challenges associated with self-centering precast piers. Two different forms of replaceable ASSPDs, one with infill steel plates perforated by elliptical holes and the other with peanut-shaped holes, are proposed. These devices are integrated into self-centering precast piers to create a novel structure that consists of multiple segments assembled with hybrid ASSPDs, connecting steel plates, and PT tendons. Experimental and numerical studies are conducted to investigate the seismic performance of the proposed self-centering precast piers. The experimental program includes details of the tests, and the results are presented and discussed in terms of key observations, hysteresis response, energy dissipation, self-centering capacity, gap opening, and PT behavior. Additionally, numerical studies explore the effects of connecting steel plates, initial PT force, and infill plate perforation geometries on the hysteresis behavior of self-centering precast piers.

2. Characteristics of SC-PSBPs

2.1. Components and mechanism of SC-PSBPs

The self-centering precast segmental bridge pier developed in this study comprises four main components: precast segments, PT tendons, connecting components, and energy dissipation components (Fig. 1). The precast segmental components primarily bear the vertical gravity load transmitted from the upper bridge deck. The PT tendons serve as both the self-centering and connecting components of the structure, establishing a "dry" PT connection between two precast components. In the novel SC-PSBP, the lateral strength is provided by the strength of the PT connection, the connecting steel plate, and the ED devices near the pier bottom. The PT connections are allowed to form a gap and rock about rocking point. The formation of this gap results in the PT tendons elongate elastically that provides the forces for self-centering of the pier. In comparison to traditional cast-in-place bridge piers, the precast segments of SC-PSBPs with proper design are expected to exhibit less overall damage at each segment end under earthquake loading. The PT connections primarily serve to recenter the pier after earthquake excitations, reducing the residual displacement and facilitating rapid post-earthquake repairs.

The incorporation of connecting steel plates at each segment joint enhances the lateral loading capacity of the SC-PSBPs. However, these connecting plates restrict the gap opening of the segment joint at large drifts due to the increased decompression moment at each joint. Additionally, the connecting steel plates, with sufficient out-of-plane stiffness and shear capacity, contribute to the sliding resistance at each segment joint. The combined effect of the shear capacity of the connecting plates and the static friction force at the joint interface provides significant resistance against applied shear forces and joint sliding at each segment joint. As a result, the connecting plates allow the entire SC-PSBP to behave similarly to an intact pier, where gap opening primarily occurs at the pier–foundation joint and sliding is not permitted at each joint interface. However, the design of the SC-PSBP differs from that of a whole pier as additional consideration is required for the shear and moment resistance of each segment joint, as well as for the strength and stiffness design of the connecting steel plates.

To provide SC-PSBPs with enhanced energy dissipation capacity, this paper employed a novel type of ED devices, i.e., the auxetic steel panel shear damper. These ED devices are placed at the pier base. Their main purpose is to provide energy dissipation and improve the bearing capacity of the structural elements. When the pier is subjected to cyclic loading, the auxetic steel dampers undergo plastic deformation, serving

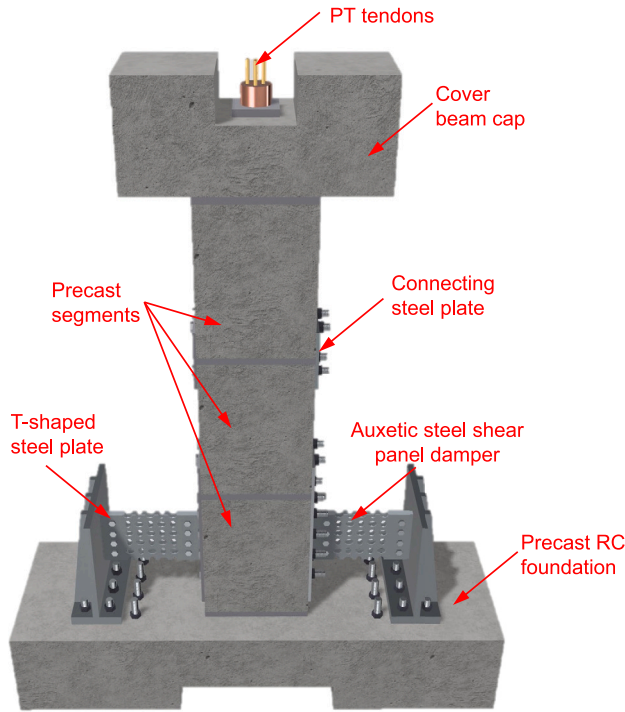


Fig. 1. Schematic illustration of the novel SC-SPBPs.

as replaceable fuses to mitigate damage to the remaining components. The inelastic deformation is confined to the ED devices, allowing for easy inspection and replacement after major earthquakes.

The proposed SC-PSBP is designed to achieve rocking mainly at the foundation interface, which allows the precast segmental pier to be considered a low-damage subsystem. Any damage is primarily limited to the pier base region, with minimal impact on the precast pier itself. The potential failure modes of the SC-PSBP include (1) concrete crushing at the pier base, and (2) fracture and complete failure of the ASSPDs. This design approach is advantageous for the seismic resilience of bridge systems, as the post-earthquake repair and recovery of SC-SPBPs only requires the replacement of the base pier segment, the ASSPD dampers, and possibly the connecting steel plates. Compared to existing self-centering precast bridge piers, the novel SC-SPBPs offer several benefits, including minimized damage to a limited number of precast segments, faster post-earthquake repair, and reduced earthquake-induced downtime and losses.

In addition to seismic performance, the proposed segmental pier offers several advantages over its existing precast counterpart, including cost, construction, and durability. One key benefit is the reduced total construction cost, attributed to the increased efficiency of concrete precast construction in an off-site prefabricating facility. Furthermore, transportation efficiency and cost are significantly reduced for the segments compared to large-scale integral bridge piers. Additionally, the on-site erection of a segmental pier proceeds more rapidly, minimizing disruptions to the existing transportation system. This in turn decreases the noise and air pollution caused by congestion and rerouting of cars during on-site construction. Although concerns regarding the durability of the proposed segmental piers may arise, it is possible to achieve a durable structure by implementing improvements in materials, construction detailing and quality assurance, as well as life-cycle maintenance [44].

2.2. Configuration and mechanism of the auxetic steel dampers

The key part of the auxetic steel dampers is essentially a perforated steel infill plate. The plate comprises unit cells that are periodically

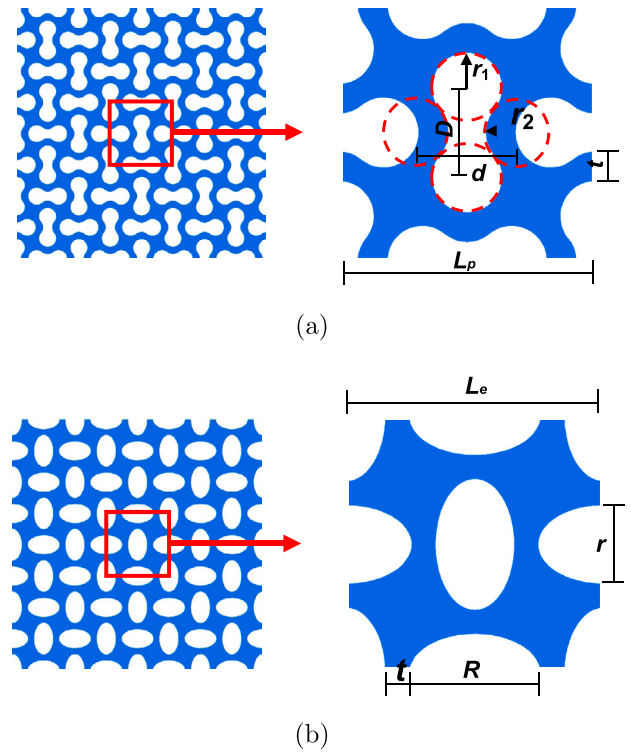


Fig. 2. Profiles of the unit cells of two perforated auxetic structures, (a) peanut-shaped perforated unit cell, (b) ellipse-shaped perforated unit cell.

arranged. Each unit cell is created by removing orthotropic cutouts. The design of two types of auxetic unit cells, namely peanut and elliptical perforation shapes, is illustrated in Fig. 2. The peanut-shaped hole is made up of two top and bottom arcs with a radius of r_1 tangent with two side arcs with a radius of r_2 . The distances between the centers of the top and bottom and the side arcs is represented by D and d , respectively. For the elliptical hole, the design parameters include the major (R) and minor axes (r). In each kind of auxetic unit cell, the minimum thickness of the ligament between neighboring holes is denoted by t . It is important to note that each unit cell representing the periodic part is square in shape and comprises a whole hole, four semi holes, and four quarter holes. The side lengths of the elliptic and peanut types of the perforated unit cells (L_e and L_p , respectively) can be expressed as:

$$L_e = R + r + 2t \quad (1)$$

$$L_p = 2(r_1 - r_2) + D + d + 2t \quad (2)$$

The porosity of each unit cell (ϕ) is computed as:

$$\phi = \frac{4A}{L^2} \quad (3)$$

where L and A represent the length of the side of each unit cell and the area of each perforated hole, respectively. Additionally, the areas of the ellipse (A_e) and the peanut (A_p) can be expressed as:

$$A_e = \pi Rr \quad (4)$$

$$A_p = 2\pi r_1^2 + \frac{Dd}{2} - 2r_2^2 \arcsin \frac{d}{2(r_1 + r_2)} - 2r_1^2 \arccos \frac{d}{2(r_1 + r_2)} \quad (5)$$

As two typical auxetic perforated structures, the elliptical and peanut-shaped perforated unit cells are well recognized to behave similarly, with four identical solid parts rotating about the ligaments between each pair of perforations [45]. The popular rotating square

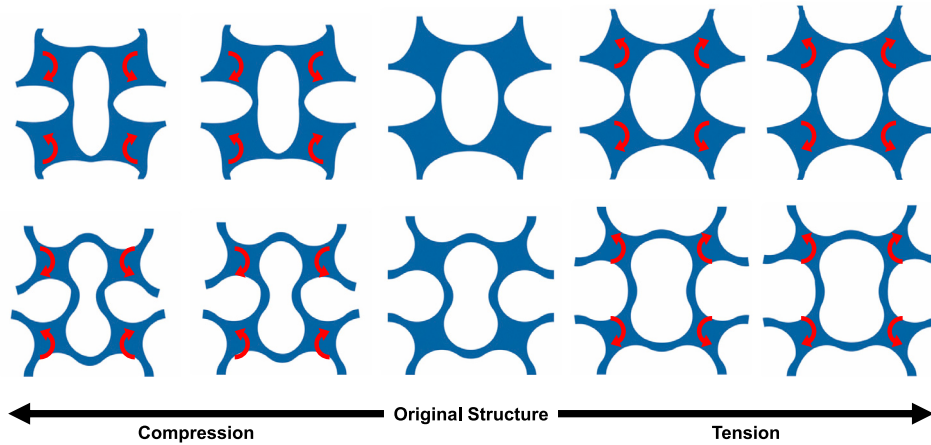


Fig. 3. Illustration of auxetic perforated unit cells subjected to uniaxial tension and compression.

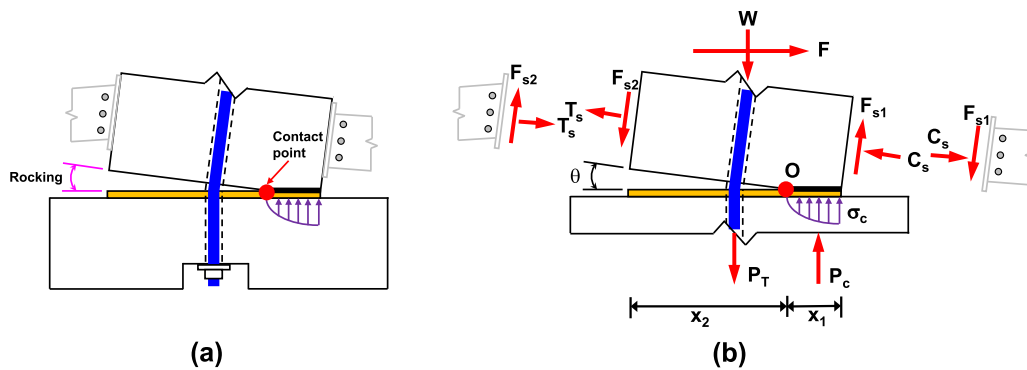


Fig. 4. Schematic illustration of deformation and free-body diagram of SC-PSBP, (a) deformation at the pier base, and (b) free-body diagram of SC-PSBP at the pier base.

(RS) model, developed by Grima and Evans [46], can be applied to both types of unit cells to model the deformation mechanism under tension and compression. In the case of uniaxial loading conditions, the unit cell deforms as the four square-like solid parts rotate, resulting in expansion and contraction of the central full perforation bidirectionally when subjected to uniaxial tension and compression, respectively (Fig. 3). The ligament separating two neighboring cutouts acts as the rotational hinge in the RS model. Consequently, the overall deformation exhibits the characteristic of a negative Poisson's ratio (auxetic).

As the ED devices are placed near the pier base, the auxetic steel dampers are subjected to complex loading conditions, particularly at the post-decompression stage. Fig. 4 shows the free-body diagram of the pier system at the state of opening of the pier-foundation interface. Using the equilibrium of moment with the contact point O gives:

$$F \cdot H = (W + P_T) \cdot \left(\frac{b - \Delta}{2} - x_1 \right) + P_C \frac{x_1}{2} + M_{d,v} \quad (6)$$

where F is the lateral force, H is height of the pier, W is the weight of the pier, P_T is PT force, b is the width of the pier cross-section, x_1 is the distance from the contact point O to the concrete compressive extreme fiber, Δ is the lateral displacement at the pier top tip, and $M_{d,v}$ is the total moment by the shear forces F_{s1} and F_{s2} , which is computed as:

$$M_{d,v} = F_{s1}x_1 + F_{s2}x_2 \quad (7)$$

The equilibrium equation of force along the vertical direction can be expressed as:

$$W + P_T + F_{s2} = P_C + F_{s1} \quad (8)$$

The shear forces F_{s1} and F_{s2} correspond to the restoring forces provided by the ASSPDs at the two sides of the pier, respectively. It

is evident that each ASSPD is subjected to combined axial and shear loads. A previous study [42] has demonstrated that auxetic perforated shear walls exhibit a stable hysteresis response without a pinching effect when subjected to cyclic shearing. This is attributed to the increased shear modulus, critical buckling load, and delayed shear buckling resulting from the auxetic effect. However, the combined loading condition in the current pier system may negatively impact the hysteresis performance of ASSPDs. In consideration of the buckling equation for the interaction of axial compression and shear loads on plates [47,48], which can be expressed as:

$$\frac{C}{C^{cr}} + \left(\frac{F_s}{F_s^{cr}} \right)^2 = 1 \quad (9)$$

where C^{cr} and F_s^{cr} are the critical buckling loads under axial compression and pure shear, respectively, and C and F_s are the in-plane compression and shear loads, respectively. It is evident from Eq. (9) that the presence of a compressive load reduces the shear load at failure in a given plate structure. However, the incorporation of a perforated auxetic configuration shows beneficial in inhibiting both shear and compression buckling. As discussed earlier, the auxetic structure under compression deforms towards densification, resulting in an enhanced critical buckling load and buckling resistance capacity. Consequently, both the quantities C^{cr} and F_s^{cr} in Eq. (9) increase, making the APSSDs more suitable as ED devices in SC-PSBPs.

3. Experimental program

3.1. Design concepts of SC-PSBP members

The design of SC-PSBP members involves considering various factors such as the lateral load capacity of the assembled pier, the PT

tendons, connecting steel plates, ED devices, the beam cap, and the RC foundation block. In the initial stage of design, it is crucial to develop a methodology for estimating the lateral strength. In the case of the SC-PSBP pier, it is idealized as a dissipative controlled rocking column [49]. Therefore, the lateral strength is primarily determined by the flexural strength at the bottom joint. The flexural capacity of the bottom joint, denoted as M_R , is a combination of several contributing factors, including the PT tendons (M_{PT}), the bottom concrete segments (M_{CC}), the ASSPD devices (M_{ED}), and the axial applied loads (M_N). The total moment resistance of the joint, computed about the centroid of the concrete compressive force, can be expressed as:

$$M_R = M_{PT} + M_N + M_{ED} \quad (10)$$

where each quantity is given as:

$$\begin{aligned} M_{PT} &= P_T \left(\frac{d - x_1}{2} \right) \\ M_N &= W \left(\frac{d - x_1}{2} \right) \\ M_{ED} &= F_{s1} \left(\frac{x_1}{2} \right) + F_{s2} \left(\frac{d - x_1}{2} \right) \end{aligned} \quad (11)$$

and calculation of the PT force requires the rocking rotation θ , which can in turn be used to derive the elongation of the PT tendons $\Delta l_p = (x_2 - d/2)\theta$. To estimate the rocking rotation, a concrete strain of $\epsilon_c = 0.003$ is considered at the extreme compressive fiber, and a simplified relationship of $\theta = 5.25\epsilon_c$ recommended by AASHTO LRFD Bridge Design Specifications [50] is adopted in this design procedure.

To achieve adequate self-centering and energy dissipation capacity, a parameter (η) quantifying the ratio of the contributions of the self-centering components to that of the ED devices is often employed, which is given as:

$$\eta = \frac{M_{PT} + M_N}{M_{ED}} \quad (12)$$

The value of η is suggested within the range of 1.15 to 1.5 for design of the bottom joint [49].

When designing a rocking pier, it is essential to ensure that the PT tendons do not yield or sustain damage. A larger initial PT force can lead to a violation of this design principle, while a smaller initial PT force can have a negative impact on the self-centering capacity. According to Ibrahim et al. [51], the axial compression ratio generated by the superstructure and the prestress axial compression ratio typically range from 0.05 to 0.1 in a typical bridge pier. Bu et al. [52] argue that in order to mitigate the significant $P-\Delta$ effect in precast segmental self-centering columns, the axial compression ratio, taking into account the contributions of both gravity and the initial PT force, should not exceed 0.3. The initial ratio of prestress axial compression for the pier can be calculated as:

$$n = \frac{P_T^0}{f'_c A_c} \quad (13)$$

where P_T^0 is the initial PT force, f'_c is the concrete compression strength, and A_c is the cross-sectional area of each segmental pier.

Design for shear of SC-PSBP requires shear resistance across the bottom and above segment joints. With respect to the bottom segment joint, the nominal shear capacity can be calculated as [50]:

$$V_n^b = C_s + T_s + 0.75(W + P_T) \quad (14)$$

Since the in-plane axial stiffness of the thin plates in ASSPDs and the in-plane axial deformation caused by pier rocking are minimal, the quantities C_s and T_s in Eq. (14) are not considered in this study. However, at segment joints other than the bottom one, measures are taken to prevent sliding at the joint interface. The shear resistance of each intermediate joint comprises contributions from friction and the connecting steel plates, denoted as V_{csp} , which can be expressed as:

$$V_n^m = \mu_c (W_a + P_T) + V_{csp} \quad (15)$$

where μ_c is the coefficient of friction at the joint interface, W_a is the weight above the given joint, $V_{csp} = 0.577 A_{csp} f_{y0}$, A_{csp} = area of the transverse cross section, and f_{y0} = yield strength of connecting steel plates. As the SC-PSBP are designed to exhibit minimal rocking at the intermediate segment joints, sufficient in-plane axial stiffness and strength of connecting steel plate are also required to provide considerable segment joint flexural capacity. To this end, the equation for designing the ASSPDs (Eq. (12)) is adopted for the connecting steel plate. The parameter η in Eq. (12) is recommended be less than 0.5 to ensure a major contribution of connecting steel plate to the total joint moment resistance.

In the design of ASSPDs, it is important to ensure both the auxetic effect and shear strength to meet the shear demands of SC-PSBP. To achieve the desired auxetic effect, the proposed perforated unit cells are designed using the method of the pattern scale factor (PSF), as developed by Ren [53] and Ren et al. [54]. The parameter PSF, which approximates the porosity of the unit cell, is suggested to have a value in the range of 40% to 60% to achieve a good auxetic effect and strength. Predicting the shear strength of the auxetic perforated shear damper directly is challenging due to a lack of well-established models specifically applicable to the perforated auxetic structures. However, existing models developed for regular perforated shear steel panels [43] can be utilized to estimate the shear strength, which can be expressed as:

$$V_{yp,asspd} = \beta(1 - \phi)V_{yp,solid} \quad (16)$$

where $V_{yp,asspd}$ and $V_{yp,solid}$ are the shear strengths of the auxetic perforated shear panel and the solid shear panel without cutouts, respectively, and β is the modification factor which is recommended within the range from 0.5 to 0.7. The shear strength of the solid shear panel [55] is expressed as:

$$V_{yp,solid} = 0.42 f_{y0} t_w L_{cf} \quad (17)$$

where t_w and L_{cf} are the thickness of ASSPD plate and the clear width of the ASSPD panel, respectively.

3.2. Details of the specimens

The experimental program comprised eight specimens, with the primary objective of investigating the effects of ASSPD geometric parameters on the cyclic behavior of SC-PSBPs. To achieve this, all parameters, except for the dampers, were kept consistent across the eight specimens. The overall dimensions of the experimental specimen is illustrated in Fig. 5. Each specimen consisted of precast cover beam cap, precast segmental piers, precast foundation, ASSPDs, PT strands, and connecting steel plates. The precast cover beam, segmental piers, and foundation were assembled through PT tendons. The precast beam cap had a cross-sectional dimension of 280×300 mm and a length of 700 mm. A cuboid groove with dimensions of $200 \times 280 \times 120$ mm was reserved in the middle for the upper anchor. The cross-sectional dimension of the three precast segmental piers was 330×280 mm, with a height of 440 mm. The precast foundation had a planar dimension of 500×300 mm and a length of 1550 mm. Similarly, a cuboid groove of $500 \times 550 \times 100$ mm was reserved in the middle of the precast foundation to accommodate the lower anchor. The geometric details and reinforcement of all the precast concrete members are also presented in Fig. 5.

Each pier segment featured a 75-mm-diameter duct for the passage of PT tendons. Additionally, multiple bolt ducts with a diameter of 22 mm were incorporated on the sides of each segment. Two additional 75-mm-diameter ducts on both sides of the precast foundation were designed for high-strength bolts with a diameter of 60 mm, which connected the foundation to the ground. Moreover, the precast foundation had six rows of pre-embedded 10.9-grade high-strength bolts, each with a diameter of 20 mm. These bolts were used to anchor the T-shaped steel plates, each consisting of two vertical steel plates

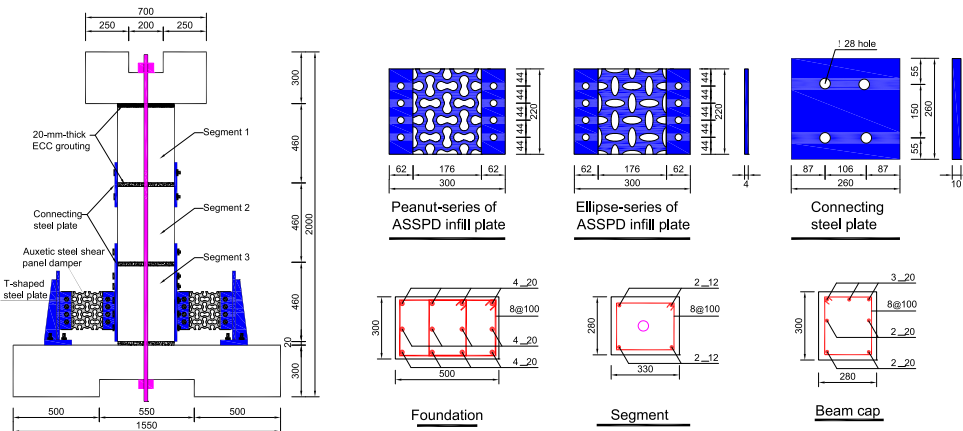


Fig. 5. Details of test specimen.

and a stiffener. To prevent local crushing of the precast pier due to force transfer from the dampers to the pier segment, two 10-mm-thick connecting steel plates were bolted to the two sides of Segment 3. Thin steel plates, welded to the steel plates, were used to connect the dampers and transfer force from the dampers to the steel plates. Two identical ASSPDs were used on each side of the pier specimen to ensure sufficient energy dissipation capacity provided by the ASSPDs. In each ASSPD, two auxetic steel infill plates were parallelly installed at the T-shaped supports and connecting steel plates.

During the tests, the PT tendons utilized were constructed using strands made up of seven wires. Each individual wire had a diameter of 15.2 mm and a cross-sectional area of 140 mm². The axial compression ratio n and the initial PT force P_T^0 in Eq. (13) were selected as 0.18 and 320 kN, respectively, for all the eight specimens. With the predefined segmental cross-section and initial PT force, the strength of APPSDs was determined using Eq. (12) with the selected value of the parameter η in the range from 1.2 to 1.5. Given the required capacity, the overall dimensions along with panel porosity of ASSPDs can be estimated using Eqs. (16) and (17). Subsequent detailed design of the auxetic perforation configuration was based on the methods discussed by Wang et al. [43]. The geometry of each connecting steel plate at the intermediate segments was designed using Eqs. (12) and (15) upon a conservatively selected value of $\eta = 0.5$. It is worth noting that the width of each connecting steel plate was simply chosen as that of the pier segment.

The design procedure mentioned above results in the geometric parameters of each ASSPD listed in Table 1, with the thickness of each auxetic steel plate of 4 mm. Two types of holes were introduced into the test specimens for a given shape of perforation. Four different porosity (ϕ) levels were achieved in the elliptical and peanut-shaped perforated dampers, i.e., $\phi = 0.54$ and 0.41 for the elliptical damper, and $\phi = 0.52$ and 0.53 for the peanut-shaped damper. The eight specimens were categorized into two groups, including one equipped with the elliptical ASSPDs (Group A) and the other with the peanut-shaped ASSPDs (Group B). Group A included three specimens with different damper configurations. Notably, the specimens Auxetic-Ellipse-2 and Auxetic-Ellipse-2-R, described in Table 2, were assembled using the components. Regarding the letter, additional out-of-plane restrainers were attached to the outer surfaces of each infill plate of ASSPD, and were connected through high-strength bolts across the two infill plates. The restrainer was essentially a steel plate with 6 mm thickness. Similarly, the specimens Auxetic-Peanut-1-R and Auxetic-Peanut-2-R in Group B had additional out-of-plane restrainers for their respective dampers. In Auxetic-Peanut-1-RR, a 20-mm-thick restrainer was filled between two infill plates, in addition to the outer restrainers. The interior restrainer was connected with the two outer restrainers and the ASSPD infill plates through bolts. Consequently, each infill plate in the ASSPDs of Auxetic-Peanut-1-RR was restrained at both sides of surfaces.

3.3. Test setup and procedure

The specimen setup is schematically illustrated in Fig. 6. The RC foundation was positioned in the first stage. The precast foundation was securely fixed to the ground using high-strength bolts (60 mm diameter) in the vertical direction, and connected to the reaction frame using PT strands in the horizontal direction. A 20-mm gap was left between each precast concrete member to accommodate a 20-mm-thick engineering cementitious composite (ECC) layer. The inclusion of the ECC layer served two purposes: (1) to prevent significant damage to the concrete at the bottom of the segmental pier, and (2) to mitigate the loss of PT force at the segmental joint caused by concrete damage. Prior to filling with ECC, four high-strength bolts were employed to secure each connecting steel plate between two adjacent precast segments. These connecting steel plates functioned as supports for the upper substructure, reducing relative rotation and lateral slip between the segments, and enhancing the overall shear capacity of the pier system. The T-shaped steel plates on both sides of the pier were affixed using high-strength bolts embedded in the foundation. At this stage, all the high-strength bolts mentioned earlier were tightened to approximately 50 kN, which is half of the target tension force.

Afterwards, the PT strands were threaded through the reserved duct and secured in place. Following that, each connection interface was grouted with ECC material. The ECC mortar was left a minimum of 72 h for curing, ensuring that the early compressive strength reached 70% of the compressive strength and surpassed the initial pre-compression due to initial PT force. Finally, the PT strands were tensioned to the target initial PT force. It is important to note that all of the PT tendons were unbonded to the precast RC members. Additionally, all of the aforementioned high-strength bolts were tightened to achieve the intended tension force (approximately 100 kN). Lastly, ASSPDs were connected to the welded steel plate using high-strength bolts as the final step in the assembly process.

3.4. Loading protocol

The actuator situated on the left side of the cover beam cap was utilized to control the lateral displacement applied at the top pier. For all experiments, a consistent loading pattern based on controlling displacement was employed. As illustrated in Fig. 7, two cycles were done at each drift level. The maximum displacements were changed in sequence of 1.55 mm, 3.10 mm, 5.425 mm, 7.750 mm, 11.625 mm, 15.500 mm, 19.375 mm, 23.250 mm, 27.125 mm, 31.000 mm, 38.750 mm, and 46.500 mm. The peak displacement (46.5 mm) at the last loading cycle corresponds to the drift level of 3%.

Table 1
Geometries of the elliptical and peanut-shaped perforated unit cells.

Specimen	Geometric dimension/mm						Non-dimensional parameter				
	r	R	t	r_1	r_2	D	ϕ	r/R	t/R	r_1/r_2	r_2/D
Ellipse-1	32.6	40.7	7.3	–	–	–	0.54	0.80	0.18	–	–
Ellipse-2	20.0	50.0	9.0	–	–	–	0.41	0.40	0.18	–	–
Peanut-1	–	–	10.1	12.1	12.1	30.7	0.53	–	–	1.00	0.39
Peanut-2	–	–	10.3	12.1	12.1	39.6	0.52	–	–	1.00	0.31

Table 2
Experimental SC-PSBP specimens.

Group	Specimen ^a	Initial PT force	ASSPD
Group A	Auxetic-Ellipse-1	320 kN	Ellipse-1
	Auxetic-Ellipse-2		Ellipse-2
	Auxetic-Ellipse-2-R		Ellipse-2
Group B	Auxetic-Peanut-1		Peanut-1
	Auxetic-Peanut-2		Peanut-2
	Auxetic-Peanut-1-R		Peanut-1
	Auxetic-Peanut-2-R		Peanut-2
	Auxetic-Peanut-1-RR		Peanut-1

^a The symbol “R” represents the ASSPDs in the specimen with out-of-plane restrainers on the outside surface of each auxetic infill steel plate, and the symbol “RR” denotes the ASSPDs with the restrainers at both sides of the steel plate.

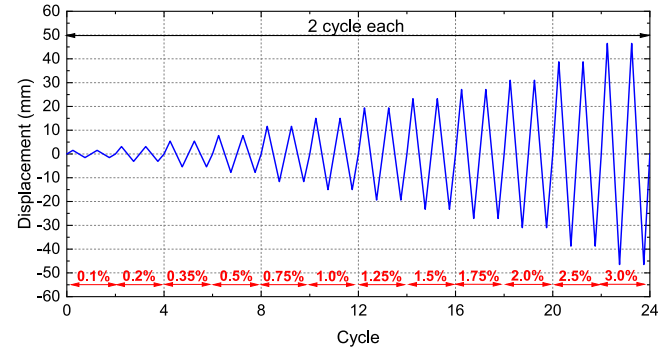


Fig. 7. Loading protocol used in the experimental tests.

displacement. The PT force was measured through a force transducer placed neighboring to the strand anchorage at the top of the cover beam. To obtain measurements of the gap open angle at each segment connection, six linear variable differential transducers (LVDT) (D2–D7) were strategically placed near the connections of the segments (Fig. 8). An additional LVDT (D8) was positioned at the foundation. The purpose was to measure the minimal horizontal movement of the foundation. The pier lateral drift was monitored directly by the test loading system.

The strain in the steel dampers were measured using four strain gauges (S1–S4) affixed to ASSPDs, specifically targeting areas where plastic straining is anticipated. In addition, four strain gauges (S5–S12) were mounted on the longitudinal reinforcing steel bars within Segment 3 to monitor the local strain in these specific steel components. Furthermore, four strain gauges (S13–S16) were positioned on the connecting steel plates.

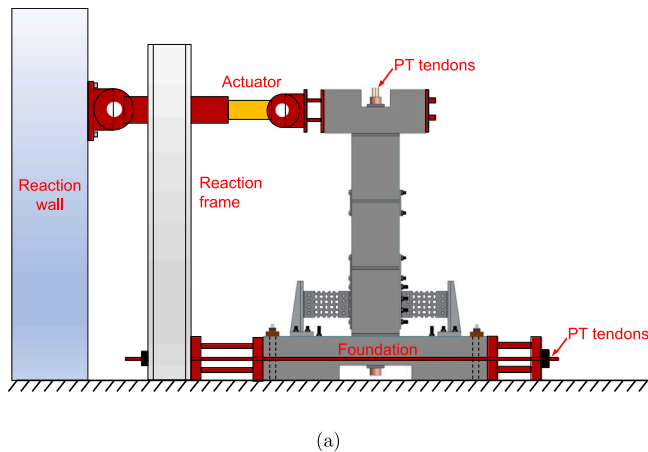
3.6. Material properties

To conduct material tests, six concrete cubic specimens measuring $150 \times 150 \times 150$ mm were prepared and cured following the guidelines provided in the Chinese standard GB/T 50081-2019 [56]. These specimens were prepared in conjunction with the precast RC segments and foundation. The compressive strength of each concrete specimen was measured and the average value was found to be 44.9 MPa. Additionally, three cubic specimens measuring $70.7 \times 70.7 \times 70.7$ mm were specifically created for material property tests on ECC. The compressive strength of the ECC was determined to be 22.3 MPa. Table 3 presents the mechanical properties of the steel reinforcements, PT strands, and steel plates.

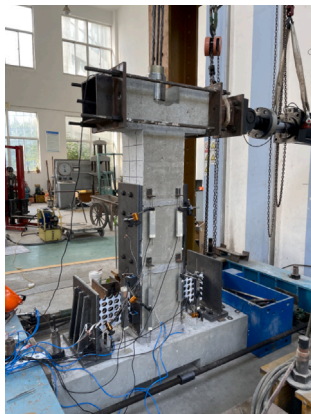
4. Experimental results and discussion

4.1. Observed behavior and failure modes

The eight specimens exhibited similar connection behavior and failure modes. During the tests, it was observed that there was invisible damage in the bottom segment due to the inclusion of connecting steel plates at Segment 3. This addition prevented the crushing of the bottom concrete. The grouted ECC layer remained intact at each connection. At the intermediate segment connections, there was minimal gap opening,



(a)



(b)

Fig. 6. Schematic illustration of test setup, (a) illustration of test setup, and (b) photograph of test setup.

3.5. Instrumentation

The test loading device consisted of a hydraulic actuator placed next to the reaction frame and the cover beam (Fig. 6). The actuator had the capability to measure and record both the lateral force and the applied

Table 3
Material properties of steels used in the experimental program.

Steel	Diameter/mm	Elastic modulus/MPa	Yield strength/MPa	Ultimate strength/MPa	Elongation (%)
HPB400	8	2.10×10^5	534	738	21.6
HPB400	12	2.00×10^5	468	627	21.4
HPB400	20	2.00×10^5	467	639	20.5
PT Strands	15.2	1.95×10^5	1720	1912	3.5
Q235	–	2.00×10^5	226	271	28.2

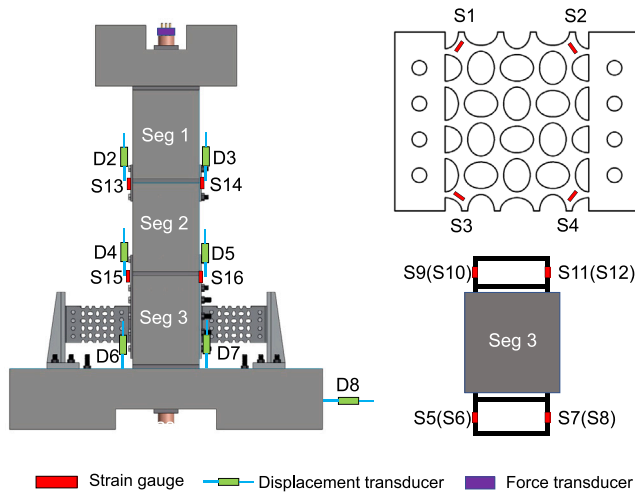


Fig. 8. Instrumentation used in the experimental tests.

indicating that the hybrid use of connecting steel plates and PT tendons effectively prevented rocking at those connections. As the lateral displacement reached approximately 15.5 mm (Drift = 1.0%), visible gap opening occurred at the bottom pier-foundation connection in each specimen. The pier did not experience any damage as the gap open angle increased. At a displacement of about 38.75 mm (Drift = 2.5%), the infill plates in the specimens of Group A displayed extensive fracture, accompanied by strength degradation in the specimens.

The failure mode of each specimen at the maximum loading drift cycle is depicted in Fig. 9. The specimens with different types of ASSPDs exhibited similar failure modes, namely, out-of-plane buckling and fracture in ASSPDs without out-of-plane restrainers. Fracture of the ASSPDs without restrainers mainly occurred at the connection between the infill plate solid edge and the unit cell ligaments. For the specimens with ASSPD out-of-plane restrainers, the out-of-plane deformation of ASSPDs was significantly mitigated.

The ASSPDs in Auxetic-Ellipse-2-R and Auxetic-Peanut-1-R demonstrated similar inward deformation. However, out-of-plane buckling was not completely eliminated in these ASSPDs. Due to their exposure to combined in-plane tension/compression and shear load states, the thin steel plates were prone to out-of-plane buckling. The use of single-sided restrainers may be insufficient in inhibiting plate out-of-plane buckling.

In contrast, the ASSPDs in Auxetic-Peanut-1-RR exhibited slight out-of-plane deformation, indicating that restrainers on both sides of the out-of-plane buckling are necessary for effectively restraining such buckling. It is worth noting that the loading process of Auxetic-Ellipse-1 was terminated at 2.5% drift due to a significant number of elliptical perforated unit cells fracturing and a drop of load in the specimen.

4.2. Hysteresis response and skeleton curve

The hysteresis curves of all the specimens exhibited the expected flag-shaped hysteresis response, as depicted in Fig. 10(a) and (b).

The hysteresis response of each specimen show pinching, suggesting that the specimens behaved to minimize residual drift after unloading. In Group A, the specimens without out-of-plane restrainers displayed superior self-centering capability compared to those with restrainers, as shown in Fig. 10(a). In Group B, the restrainers installed at the outer side of the dampers had a negligible impact on the residual drift of the specimens (Fig. 10(b)). However, Auxetic-Peanut-1-RR with both-sided restrained ASSPDs demonstrated the largest residual displacement among the specimens in the same group.

Fig. 10(c) and (d) illustrate the skeleton curves of the eight specimens. It was observed that the specimens in Group A experienced fracture at the solid matrix of the ASSPD infill plates at 2.0% drift, resulting in a significant decrease in load-carrying capacity. Although the additional out-of-plane buckling restrainers contributed to an improvement in the ultimate strength of the Group A specimen, strength degradation still appeared at 3% drift. A comparison between Auxetic-Ellipse-1 and Auxetic-Ellipse-2 revealed that the ASSPD devices had a significant influence on the ultimate strength of SC-PSBPs. The higher ultimate strength of Auxetic-Ellipse-2 was a result of the lower porosity of the infill plate utilized in the ASSPDs. Conversely, the specimens in Group B showed a continuous increase in the load-bearing capacity. The strength of Auxetic-Peanut-1 at 3% drift was 14% higher than that of Auxetic-Peanut-2. Although the ASSPDs installed in these two specimens had the infill plates with similar porosities, the geometric configuration of each unit cell in the ASSPDs differed. The presence of out-of-plane buckling restrainers had an insignificant impact on the strength enhancement of the SC-PSBPs in Group B.

A comparison of the skeleton curves between the specimens in the two groups reveals significant differences in their load-displacement response. These differences can be attributed to the peanut-shaped ASSPDs, which possess good in-plane deformation capacity under combined in-plane tension/compression and shearing. This finding is supported by experimental observations, which indicate that the peanut-shaped ASSPDs did not rupture significantly throughout the loading process. Despite both types of ASSPDs having the infill plates with comparable porosities, the cyclic response demonstrated notable distinctions. This underscores the importance of an optimal design for the perforation shape and geometric configuration of a unit cell in auxetic structures.

Table 4 summarizes the forces and displacements of each specimen at the yield and ultimate points. Additionally, the displacement ductility of each specimen is also included in Table 4. To determine the yield and ultimate points of the eight specimens, the equivalent energy elastic-plastic (EEEP) method, as specified in ASTM E2126-11 [57], was employed. The failure point of each specimen is defined as the instance when the strength drops to 80% of the peak load. The displacement ductility μ_d is quantified by the ratio of the displacement at failure (Δ_u) to that at yielding (Δ_y), which is given as:

$$\mu_d = \frac{\Delta_u}{\Delta_y} \quad (18)$$

The specimens in Group B exhibited no strength degradation throughout the entire tests. As a result, the displacements at failure for those specimens exceeded those at the termination of the test. Generally, the ductility coefficients of all specimens exceeded 3.0. Furthermore, compared to those in Group B, the specimens in Group A demonstrated higher ductility coefficients. The ductility of the specimens can be strongly correlated to the deformation capacity of ASSPDs.

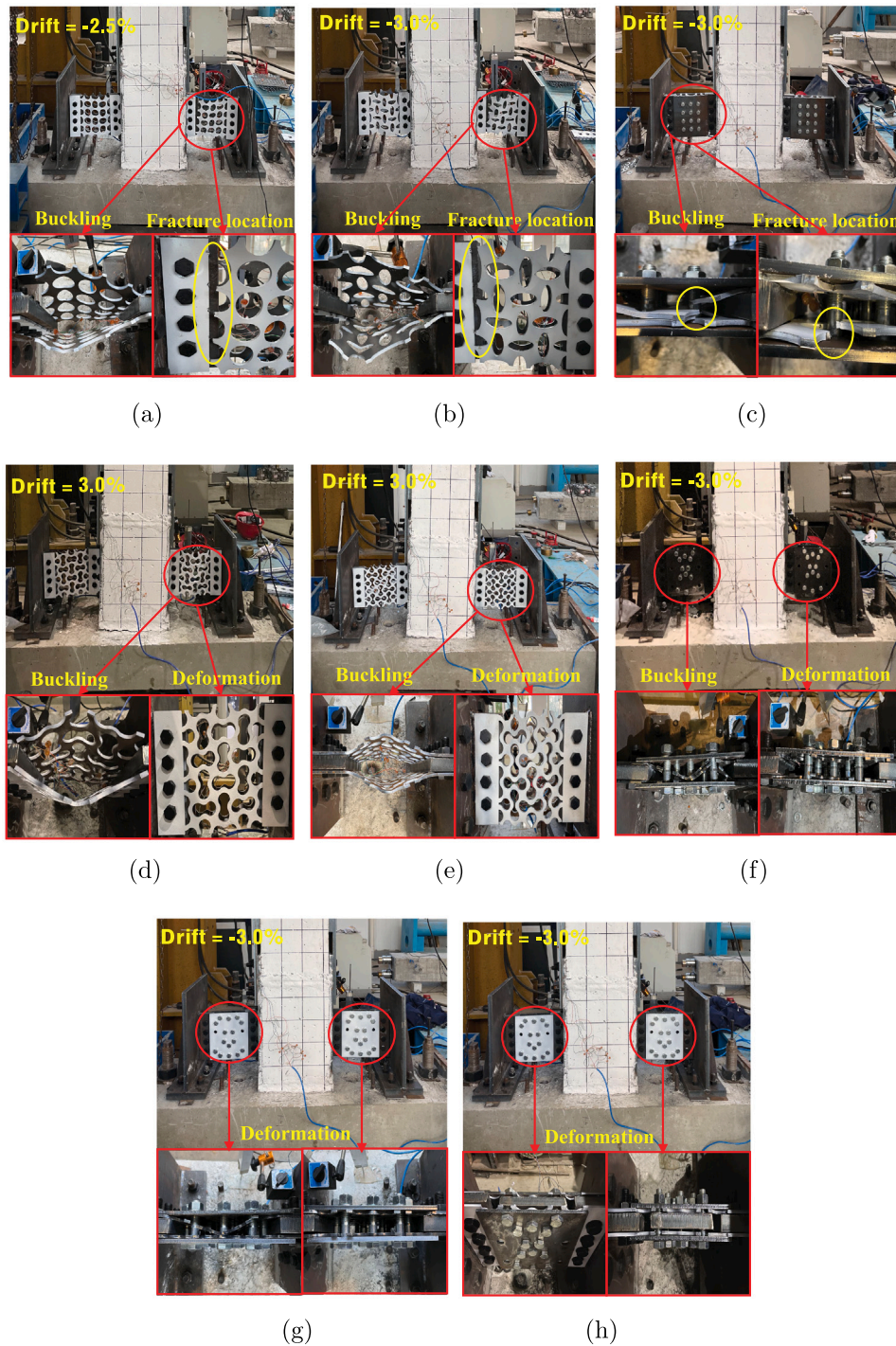


Fig. 9. Failure mode of each specimen, (a) Auxetic-Ellipse-1, (b) Auxetic-Ellipse-2, (c) Auxetic-Ellipse-2-R, (d) Auxetic-Peanut-1, (e) Auxetic-Peanut-2, (f) Auxetic-Peanut-1-R, (g) Auxetic-Peanut-2-R, and (h) Auxetic-Peanut-1-RR.

4.3. Stiffness degradation

The secant stiffness (K_{sec}) is determined by calculating the slope of the line connecting the highest and lowest loading points in both the positive and negative directions of the hysteresis curve during a specific loading cycle (Fig. 11). It can be calculated as:

$$K_{sec} = \frac{|P_{max}^+| + |P_{max}^-|}{|\Delta_{max}^+| + |\Delta_{max}^-|} \quad (19)$$

where P_{max}^+ and P_{max}^- represent the maximum forces in the positive half cycle and negative half cycle, respectively, while Δ_{max}^+ and Δ_{max}^- denote the corresponding lateral displacements. Fig. 11 provides an example that demonstrates the key parameters P_{max}^+ , P_{max}^- , Δ_{max}^+ , and Δ_{max}^- , as well as the secant stiffness. This example is extracted from the hysteresis curve of Auxetic-Peanut-1 at the first loading cycle with a drift of 3.0%.

Fig. 12 displays the stiffness degradation curves of eight specimens. These specimens exhibit a similar trend of variation in the secant

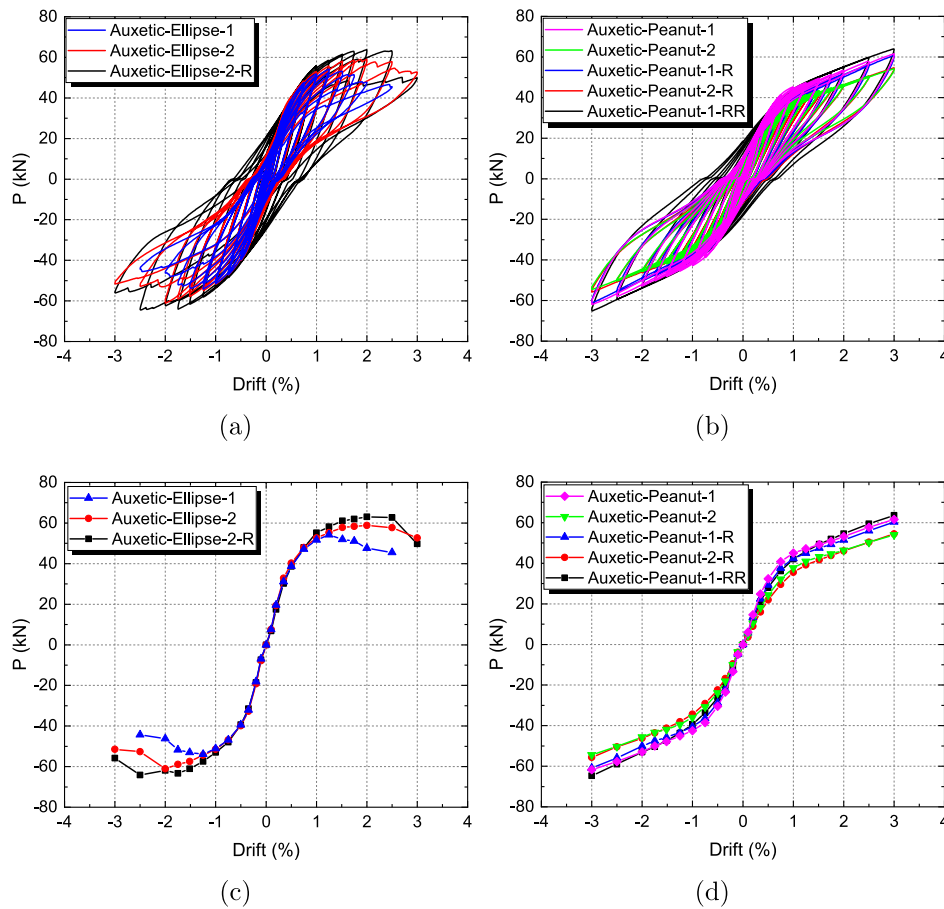


Fig. 10. Hysteresis and skeleton curves of SC-PSBP specimens, (a) hysteresis curves of specimens in Group A, (b) hysteresis curves of specimens in Group B, (c) skeleton curves of specimens in Group A, and (d) skeleton curves of specimens in Group B.

Table 4
Summary of the test results.

Specimen	Loading direction	P_y /kN	Δ_y /mm	P_u /kN	Δ_u /mm	μ_A
Auxetic-Ellipse-1	+	48.7	7.9	54.2	36.6	4.6
Auxetic-Ellipse-1	-	49.3	8.4	53.9	32.7	3.9
Auxetic-Ellipse-2	+	54.6	8.9	58.8	46.5	5.2
Auxetic-Ellipse-2	-	53.6	8.8	61.1	43.9	5.0
Auxetic-Ellipse-2-R	+	58.0	10.4	63.1	44.2	4.2
Auxetic-Ellipse-2-R	-	58.0	9.9	64.1	46.5	4.7
Auxetic-Peanut-1	+	51.2	11.2	>61.6	>46.5	>4.2
Auxetic-Peanut-1	-	50.4	11.9	>61.7	>46.5	>4.0
Auxetic-Peanut-2	+	45.1	14.0	>54.0	>46.5	>3.3
Auxetic-Peanut-2	-	44.3	14.1	>54.4	>46.5	>3.3
Auxetic-Peanut-1-R	+	49.7	12.4	>60.4	>46.5	>3.8
Auxetic-Peanut-1-R	-	48.6	12.4	>60.9	>46.5	>3.8
Auxetic-Peanut-2-R	+	45.0	15.6	>54.5	>46.5	>3.0
Auxetic-Peanut-2-R	-	44.6	15.4	>55.6	>46.5	>3.0
Auxetic-Peanut-1-RR	+	52.4	14.2	>63.6	>46.5	>3.3
Auxetic-Peanut-1-RR	-	51.5	15.2	>64.7	>46.5	>3.1

Note: P_y and P_u are the nominal yield strength and ultimate strength, respectively, and Δ_y and Δ_u are the lateral displacements corresponding to nominal yield strength and specimen failure, respectively.

stiffness. Initially, the secant stiffness rapidly decreased up to a drift of 1.25%, after which it decreased more gradually. It can be observed that the secant stiffness of Auxetic-Ellipse-1 decreased more rapidly than Auxetic-Ellipse-2 after 1.25% drift cycles. This suggests that an increase in the porosity of the ASSPD infill plate negatively affects the stiffness of both the ASSPDs and SC-PSBPs. Furthermore, the geometric parameters of the unit cell also have a significant impact on the device stiffness, even in cases with similar porosities.

In terms of the specimens in Group B, the stiffness of Auxetic-Peanut-1 at 0.2% drift was 33.6% greater than that of Auxetic-Peanut-2 (Fig. 12(b)). The ASSPDs of both specimens had similar porosities and the same magnitude of R/r . The notably higher r_2/D value in the former specimen contributed to the overall stiffness improvement. Additionally, the stiffness of SC-PSBPs in Group B, which included ASSPD out-of-plane buckling restrainers, was smaller compared to the counterpart without such restrainers, up to 2.0% drift cycles. However,

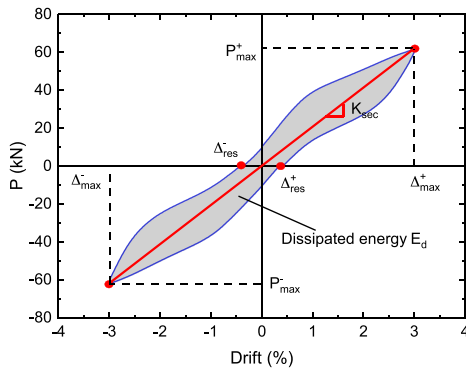


Fig. 11. Illustration of secant stiffness using a representative hysteresis loop of Auxetic-Peanut-1 at 3% drift.

from that point onwards, the stiffness had an insignificant difference among specimens with the same ASSPDs.

4.4. Dissipated energy and equivalent viscous damping

The energy dissipation capacity of the specimen is assessed using the total dissipated energy (E_d), which is defined as the area enclosed by a full hysteresis loop [11]. In addition, the equivalent viscous damping ratio (ξ_{eq}) is adopted to analyze the influence of the ASSPD geometric parameters on the energy dissipation capability of SC-PSBP. The equivalent viscous damping ratio at a given drift ratio is calculated as [58]:

$$\xi_{eq} = \frac{E_d}{2\pi K_{sec} \Delta_0^2} \quad (20)$$

where $\Delta_0 = (|\Delta_{max}^+| + |\Delta_{max}^-|)/2$.

Each specimen exhibited an increase in E_d with drift during the initial loading phase (Fig. 13(a) and (b)). The specimens in Group A demonstrates a decrease in the dissipated energy at larger drifts. This degradation in energy dissipation capacity of SC-PSBP can be attributed to significant out-of-plane buckling, plastic deformation, and fracture failure in the ASSPD infill plates, which served as the primary energy dissipators in SC-PSBPs. Generally, the inclusion of out-of-plane buckling restrainers had a positive effect on the energy dissipation capacity of the specimen. The amount of dissipated energy of Auxetic-Ellipse-2-R was 71% higher than that of Auxetic-Ellipse-2 at 2.5% drift. It is important to note that although the porosity of ASSPDs in Auxetic-Ellipse-1 was small, the energy dissipation capacity decreased at 1.75%, which was the earliest among the three specimens in Group A. It can be observed that the specimens in Group B exhibited a consistent increase in energy dissipation capacity with drift, and Auxetic-Peanut-1-RR with internal and external buckling restrainers showed superior energy dissipation compared to the other specimens in Group B.

The equivalent viscous damping ratio of each specimen is illustrated in Fig. 13(c) and (d). It can be observed that all specimens exhibited the peak ξ_{eq} within the range of 0.1 to 0.2. All specimens, except for Auxetic-Peanut-2-R and Auxetic-Peanut-1-RR, showed visible degradation in ξ_{eq} . The incorporation of ASSPD out-of-plane buckling restrainers contributed to delaying the dropping of ξ_{eq} . However, it was found that only the double-sided ASSPD out-of-plane buckling restrainers effectively enhanced the equivalent viscous damping ratio. At 3.0% drift, the viscous damping coefficient of Auxetic-Peanut-2 was 16% lower than that of Auxetic-Peanut-1. This highlights the significant impact of perforation shape of ASSPDs on the viscous damping properties of SC-PSBP specimens.

4.5. Re-centering capacity

The relative self-centering coefficient (RSE) is introduced to assess the self-centering capacity of SC-PSBPs. The coefficient is a measure of the amount of peak displacement that is recovered upon unloading [28], which is expressed as:

$$RSE = 1 - \frac{|\Delta_{res}^+| + |\Delta_{res}^-|}{|\Delta_{max}^+| + |\Delta_{max}^-|} \quad (21)$$

where Δ_{res}^+ and Δ_{res}^- correspond to the displacements at $P = 0$ in the positive and negative loading directions during a particular drift cycle, respectively. It is clear that a larger RSE corresponds to a greater re-centering capacity of SC-PSBP.

Fig. 14(a) and (b) illustrate the variation of residual displacement in the eight specimens. The residual displacement in each specimen is primarily a result of accumulated plastic deformation in ASSPDs. In the specimens of Group A, the residual displacements initially increased during early loading periods and then decreased at larger drifts. For Auxetic-Ellipse-1 and Auxetic-Ellipse-2, the maximum residual displacements were observed at 1.25% and 1.5% drift, respectively, reaching magnitudes of 4.4 mm and 5.5 mm. Notably, the installation of buckling restrainers significantly increased the residual displacement. In fact, the residual displacement of Auxetic-Ellipse-2-R at 1.75% drift was 115.6% larger than that of Auxetic-Ellipse-2. The accumulation of plastic deformation in ASSPDs contributed to an increment in the residual displacement, while the fracture of infill steel plates in ASSPDs at large drifts may have caused a decrease in residual displacement. In Group B specimens, the residual displacement increased linearly with drift throughout the loading history. All specimens in Group B exhibited their maximum residual displacement at the end of testing. Compared to Auxetic-Peanut-1, Auxetic-Peanut-1-RR had a residual displacement that was 91.3% greater at maximum drift. This phenomenon is similar to that observed in the pair of Auxetic-Ellipse-2 and Auxetic-Ellipse-2-R. The possible explanation for this is that the out-of-plane buckling-restrainers constrained the out-of-plane deformation of each ASSPD, inducing more accumulative in-plane plastic deformation.

Fig. 14(c) and (d) present the results RSE versus drift in each specimen. With the exception of Auxetic-Ellipse-2-R, all specimens exhibited RSE values ranging from 0.7 to 0.9, indicating satisfactory self-centering capacity. The RSE value of each Group A specimen decreased at drifts less than 1.0% and then gradually increased as fractures occurred in the infill steel plates. This partial failure of ASSPDs led to a reduction in lateral load capacity for resisting specimen re-centering. In contrast, Group B specimens displayed relatively stable RSE values throughout the loading history. Notably, the average RSE values for Auxetic-Peanut-1 and Auxetic-Peanut-2 remained above 0.8. However, the specimen Auxetic-Peanut-1-RR, with strong ASSPD restrainers, had an RSE value below 0.8. This may be attributed to the additional lateral load resistance resulting from the in-plane strength of ASSPDs during unloading.

4.6. Gap opening behavior and compression zone

The displacement drift at the top of the pier specimen (θ_w) is composed of the elastic bending deformation of each precast bridge pier segment and the gap open angles at each segment and pier-foundation connection. The gap angle at each segment connection can be calculated based on the relative displacement measured by displacement transducers during the tests:

$$\theta = \frac{|D_e| + |D_w|}{h} \quad (22)$$

where D_e represents the displacement at the left side of each connection measured by displacement transducers D2, D4, and D6, D_w represents the displacement measured by displacement transducers D3, D5, and D7, and h is the cross-sectional width of the pier.

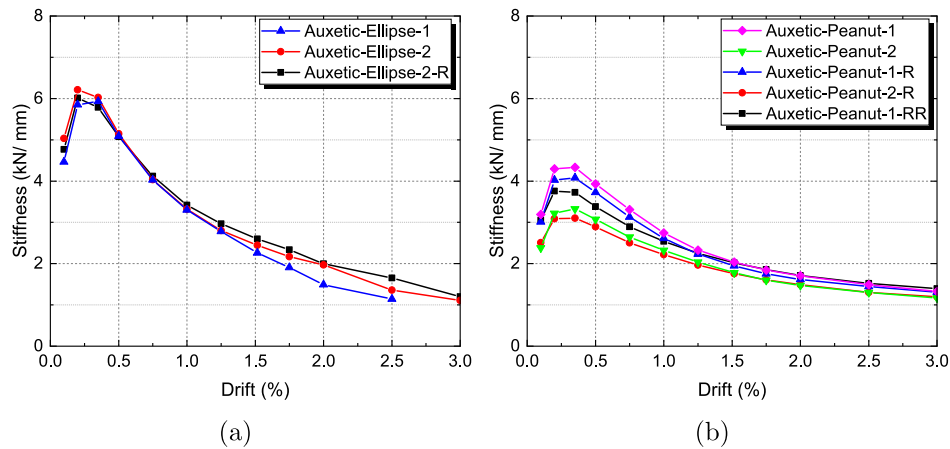


Fig. 12. Secant stiffness vs. drift, (a) specimens in Group A, and (b) specimens in Group B.

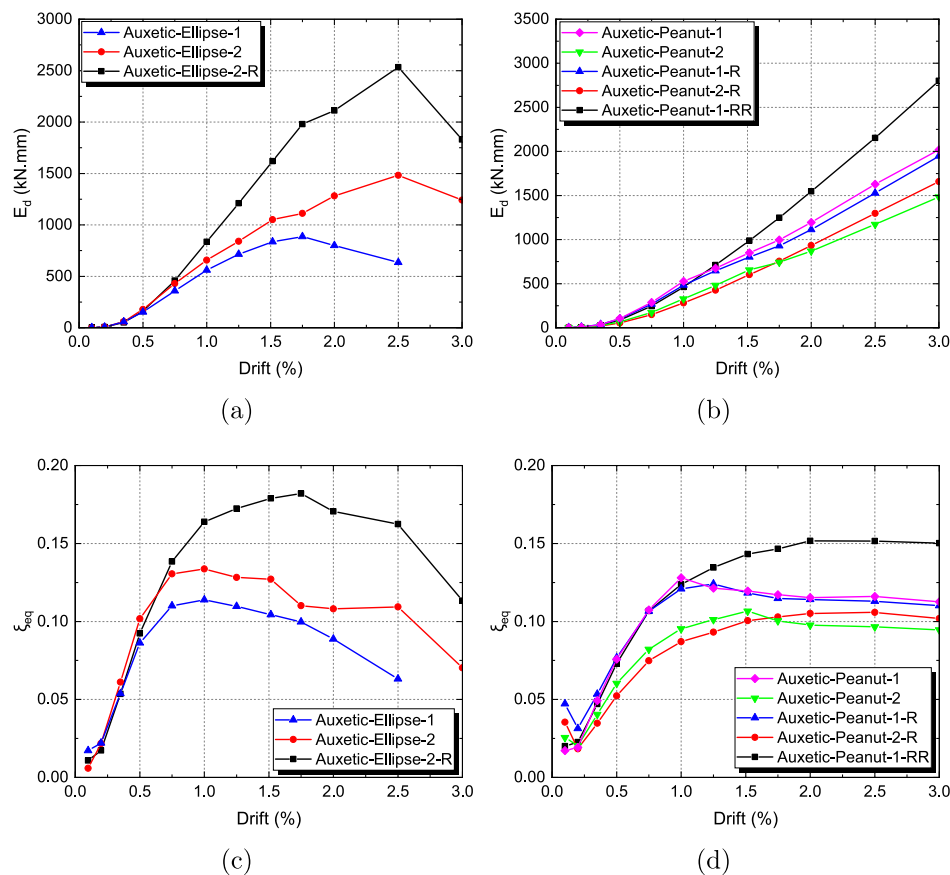


Fig. 13. Results of energy dissipation, (a) dissipated energy of Group A specimens, (b) dissipated energy of Group B specimens, (c) equivalent viscous damping ratios of Group A specimens, and (d) equivalent viscous damping ratios of Group B specimens.

The gap angles varying with drift for Auxetic-Ellipse-2 and Auxetic-Peanut-1 are presented in Fig. 15. The gap angles are categorized as those at the pier-foundation connection and at the connection of each two neighboring segments. Based on experimental observations, it is noted that the gap angle was minimal at the interface of each two neighboring segments. Therefore, the total amount of gap angles at the interfaces of Segment 1–2 and 2–3 is presented in Fig. 15. The maximum magnitudes of the total gap angles for the inner segments in Auxetic-Ellipse-2 and Auxetic-Peanut-1 were only 0.32% and 0.19%, respectively, at 3% drift. In contrast, the maximum gap angle at the pier-foundation connection was 2.26% and 2.07% for Auxetic-Ellipse-1 and Auxetic-Peanut-1, respectively. It can be concluded that the

proportion of gap open at the segment interface to the total drift of the pier was insignificant, as compared to the gap opening at the pier base.

Initially before the pier-foundation connection decompresses, the pier-foundation connection at the base has an initial stiffness equal to that of a conventional moment-resisting connection. As the lateral displacement of the specimen increases, the interface between the pier and foundation starts to open and close. There is a rotational center at the cross-section of the pier at the base, about which the pier rocks. Additionally, the compression zone of the pier cross-section near the pier-foundation connection significantly affects the behavior of the PT connection, is crucial for SC-PSBP design. The depth of the compression

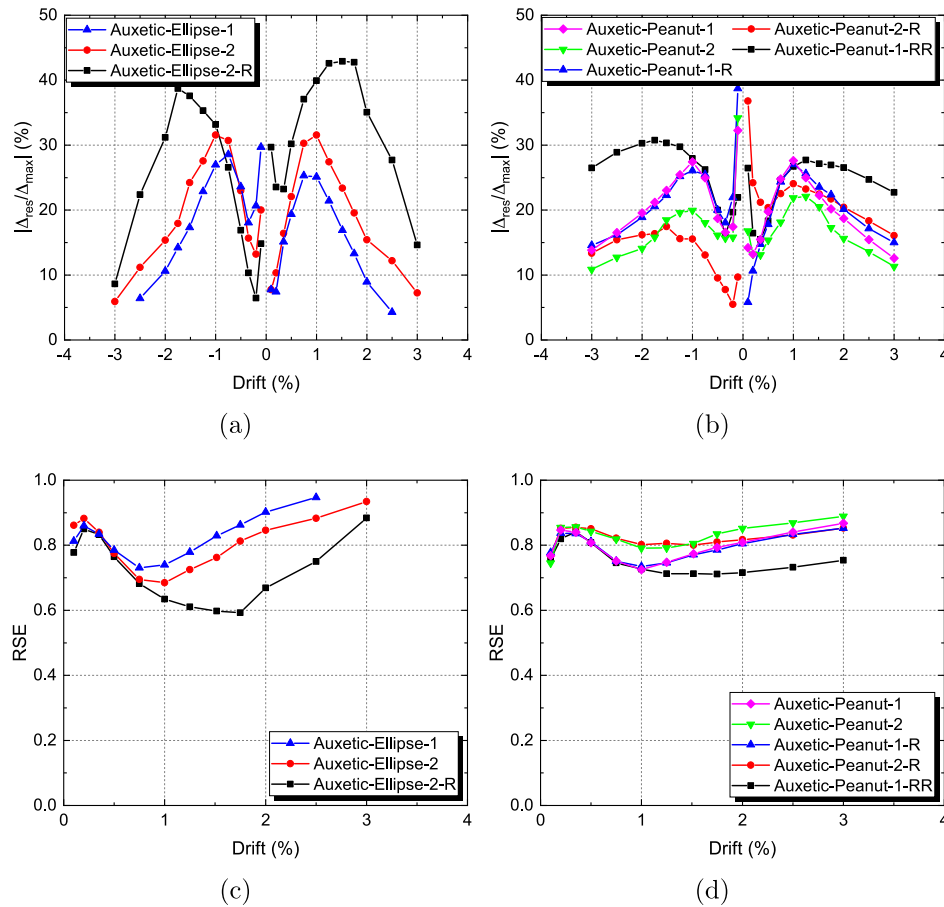


Fig. 14. Residual displacement response, (a) residual displacements of Group A specimens, (b) residual displacements of Group B specimens, (c) relative self-centering coefficients of Group A specimens, and (d) relative self-centering coefficients of Group B specimens.

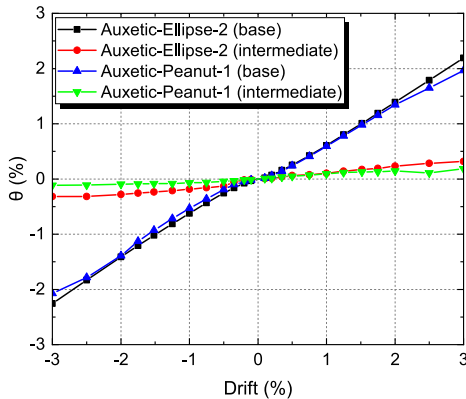


Fig. 15. Gap angles at the base and at the intermediate segments (the amount of angle at the intermediate segments is the total values of which at Segment 2–3 and 1–2 connections).

zone (h_a) can be determined from the relative displacements measured by DTs at the bottom of the pier:

$$h_a = \frac{|D_g|}{|D_g| + |D_c|} h \quad (23)$$

where D_g signifies the gap open displacement, and D_c denotes the displacement measured at the compression side of the connection. These measurements are obtained using displacement transducers D6 and D7.

Fig. 16 displays the relative depth of the compression zone (h_a/h) at the base of the pier specimen. At the initial stage, the value of h_a/h experienced a substantial decrease, going from 1 to approximately 0.3. There appeared slight changes in h_a/h with drift from 0.2% to 3% (the end of the test). The eight specimens showed similar variation of h_a/h with drift.

4.7. PT tendon behavior

The PT forces were measured using force transducers positioned near the anchors, as shown in Fig. 8. The variation of PT force with drift in Auxetic-Ellipse-1 and Auxetic-Peanut-1 is illustrated in Fig. 17. It is observed that the PT force remained relatively constant prior to the opening of the gap, corresponding to a drift lower than 0.35%. Once the gap opened, the PT force exhibited a nearly linear increase with drift, which was attributed to the elongation of the PT strands resulting from the gap opening. Overall, the loss in PT force was relatively minimal in both specimens, i.e., 4.8% and 4.4% of the initial PT force for Auxetic-Ellipse-1 and Auxetic-Peanut-1, respectively. This loss of prestress was primarily caused by the seating of the PT strands anchorage. The maximum PT force (P_T^{max}) reached at 3% drift were 414.8 kN and 409.2 kN in Auxetic-Ellipse-1 and Auxetic-Peanut-1, respectively. Notably, these values remained below the yield force of the PT strands.

4.8. Local strain response

The local strain response of the steel components was monitored in this study. Fig. 18(a) depicts the strain history of the longitudinal reinforcements at the top and bottom of Segment 3 of Auxetic-Peanut-1. It is evident that all the reinforcing steel rebars exhibited elastic

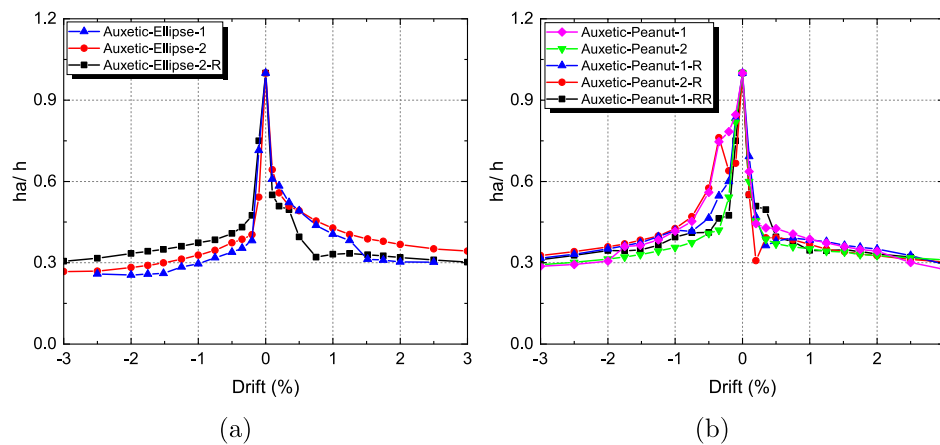


Fig. 16. Relative depths of the compression zone, (a) specimens of Group A, and (b) specimens of Group B.

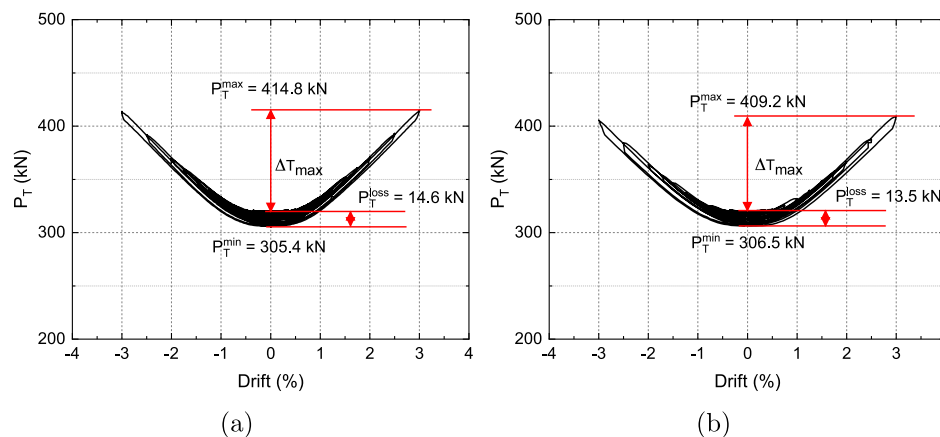


Fig. 17. PT force vs. drift, (a) Auxetic-Ellipse-1, and (b) Auxetic-Peanut-1.

behavior throughout the loading process, below the yield strain level of approximately $2500 \mu\epsilon$ (Fig. 18(b)). This strain response aligns with the experimental observations that Segment 3 did not exhibit any visible damage. In contrast, the ASSPD infill plates experienced significant straining, surpassing the yield strain at an early stage of around 1% drift. Of interest is the observation that the strain at the location of S1 was consistently lower than the yield strain throughout the loading process. One possible explanation is that the strain gauge S1 was capable of measuring strain in only one direction, while the strain in the infill plate was under a triaxial state. Furthermore, it is essential to gather strain data from the connecting steel plates. The results presented in Fig. 18(c) demonstrate that all the monitored steel plates deformed elastically. The relatively small magnitudes of strain suggest that the connecting steel plates contributed to lateral stiffness and load-bearing capacity, but did not significantly contribute to the overall energy dissipation of the SC-PSBPs.

5. Numerical simulation

5.1. Finite element model

In order to gain more in-depth understanding of the seismic performance of SC-PSBPs, a detailed three-dimension (3D) finite element (FE) model of the specimen of Auxetic-Peanut-1 is developed based on the commercially available FE platform ABAQUS [59], and the details of the FE model are presented in this section.

The developed FE model of Auxetic-Peanut-1 is depicted in Fig. 20. This model comprises various components, including the precast cover

beam, precast pier segments, RC foundation, ASSPD devices, PT strands, connecting thin steel plates, and high-strength bolts. To accurately represent these components, apart from ASSPDs and steel reinforcements, eight-node 3D solid elements with reduced integration point (C3D8R) provided by ABAQUS are utilized. The infill plates of ASSPDs are simulated using a four-node shell element with reduced integration point (S4R). The steel reinforcements are modeled using truss elements (T3D2). Mesh sizes are selected through mesh sensitivity studies to strike a balance between numerical accuracy and efficiency (Fig. 19). As shown in Fig. 19(a), the skeleton curves of Auxetic-Peanut-1 are almost coincident among the models with the precast RC components meshed by elements with sizes of 20 mm, 30 mm, and 40 mm. Similar tendency is observed in Fig. 19(b), which depicts the results of mesh sensitivity studies on the mesh refinement of connecting steel plates and ASSPDs with sizes of 2 mm, 3 mm, and 4 mm. Considering the significance of each component on the seismic response of SC-PSBPs, precast segment piers are thus discretized into elements of 20 mm size, while the beam cap and RC foundation are modeled using elements of 30 mm size. The connecting steel plates, high-strength bolts, and ASSPDs are represented by fine meshes with a size of 2 mm.

Two rigid pads are placed on top of the cover beam cap and at the bottom of the foundation to simulate anchors, and the PT tendons are connected to these rigid pads. Initial PT stress is applied to the tendons using the “cooling method”. The “tie” constraints are applied to the adjacent surfaces in the ASSPDs and the connecting steel plates. Surface-to-surface contact is defined associated with a hard contact and Coulomb friction model for the following interfaces: pier and RC

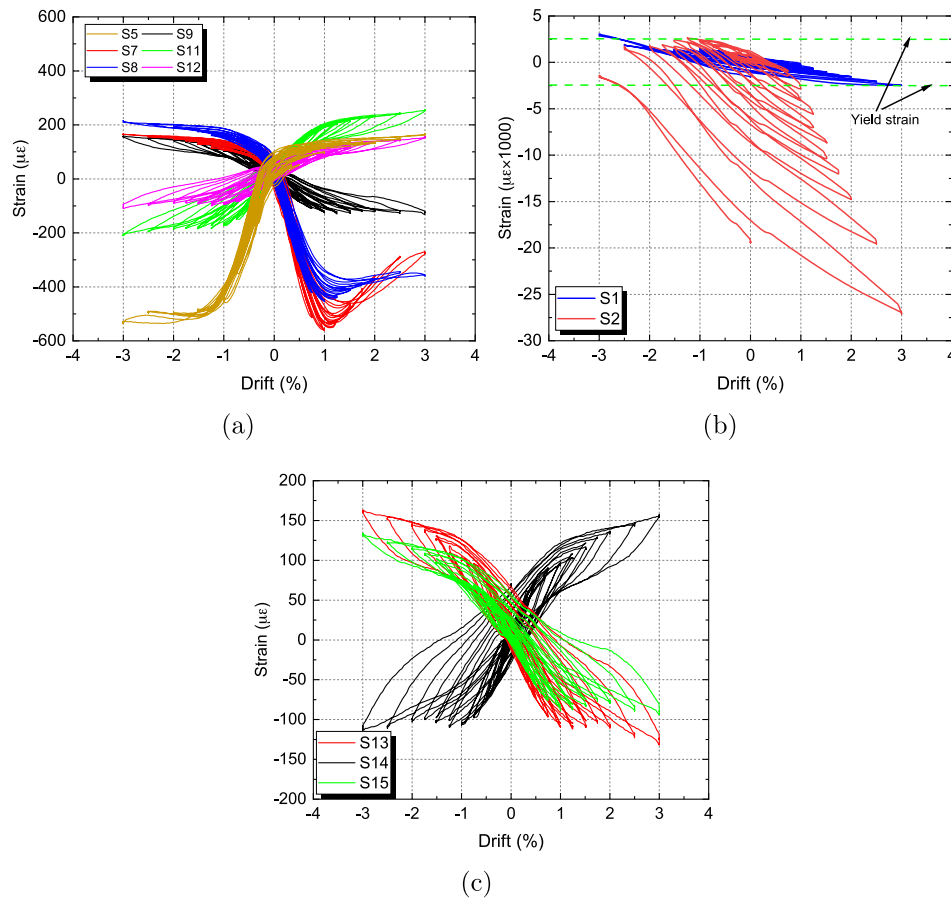


Fig. 18. Strain response of Auxetic-Peanut-1, (a) longitudinal reinforcements at Segment 3, (b) ASSPD infill plate, and (c) connecting steel plates.

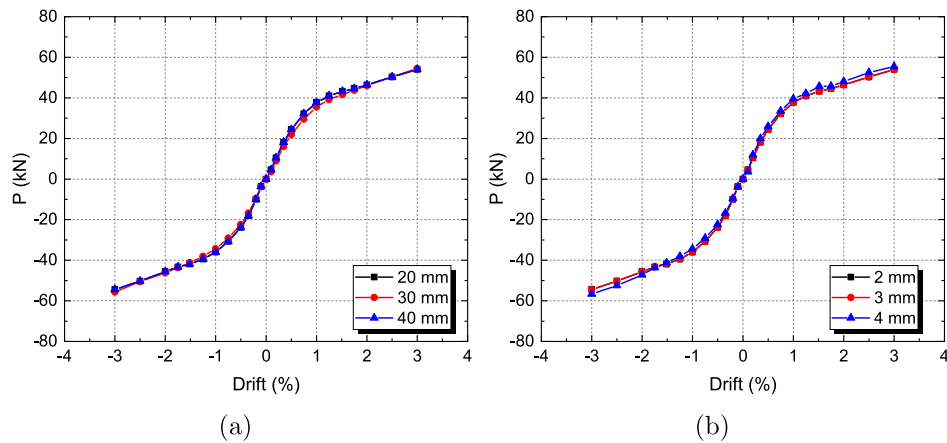


Fig. 19. Results of mesh sensitivity studies on Auxetic-Peanut-1 (a) effect of the element size of RC components, (b) effect of the element size of connecting steel plates and ASSPDs.

foundation, contacted surfaces of adjacent segments, connecting steel plates and concrete, and bolt shanks and steel plates. A friction coefficient of 0.33 is chosen for the tangent direction between the contact surfaces [60]. It is worth noting that the interaction between the steel reinforcements and concrete is simply modeled using the “embedded” constraint. A reference point located at the left side of the beam cap is established and coupled with the entire side surface of the cap. The reference point is constrained at the rotational degrees of freedom and out-of-plane translational degree of freedom. The lateral cyclic loads applied to the specimen are directly imposed on the reference point. Fixed constraints are imposed at the bottom of the RC foundation.

The inelastic behavior of concrete is represented by the “concrete damaged plasticity” (CDP) model in ABAQUS. The CDP model involves isotropic tensile and compressive plasticity and damaged elasticity. The compressive stress–strain relationship proposed by Hognestad [61] is introduced as:

$$f_c = \begin{cases} f'_c \left[\frac{2\epsilon_c}{\epsilon_{c0}} - \left(\frac{\epsilon_c}{\epsilon_{c0}} \right)^2 \right] & 0 \leq \epsilon_c \leq \epsilon_{c0} \\ f'_c \left[1 - 0.15 \left(\frac{\epsilon_c - \epsilon_{c0}}{\epsilon_{cu} - \epsilon_{c0}} \right)^2 \right] & \epsilon_{c0} \leq \epsilon_c \leq \epsilon_{cu} \end{cases} \quad (24)$$

Table 5
Material model parameters of concrete used in the finite element model.

f'_c/MPa	ϵ_{c0}	ϵ_{cu}	f'_t/MPa	ϵ_{t0}	E_c/MPa	ψ	e	f_{b0}/f_{c0}	K_c	μ
44.90	0.002	0.003	3.80	1.64×10^{-4}	3.15×10^4	30°	0.10	0.0001	1.16	0.67

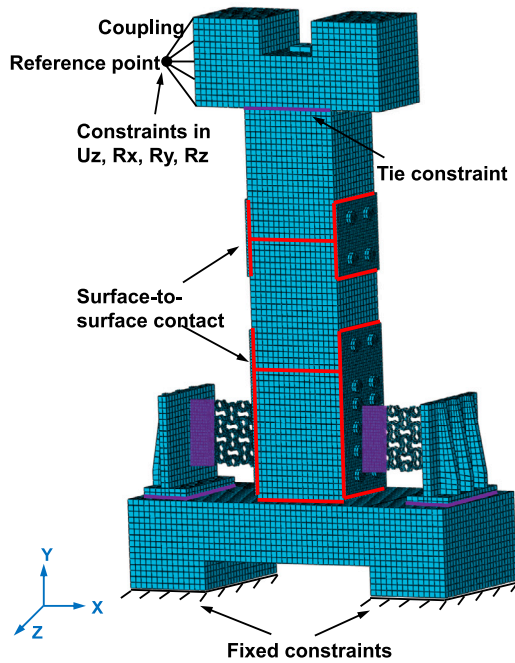


Fig. 20. Finite element model of Auxetic-Peanut-1 specimen (symbols in red and purple denote the surface-to-surface contact and “tie” constraints, respectively).

where f_c and ϵ_c are the compressive stress and strain of concrete, respectively, f'_c is the peak compressive strength at compression strain ϵ_{c0} , and ϵ_{cu} is the compressive strain of concrete at failure. The stress-strain relationship of concrete in tension developed by Attard and Setunge [62] is given as:

$$f_t = \begin{cases} f'_t \left[1.2 \frac{\epsilon_t}{\epsilon_{t0}} - 0.2 \left(\frac{\epsilon_t}{\epsilon_{t0}} \right)^6 \right] & 0 \leq \epsilon_t \leq \epsilon_{t0} \\ f'_t \left[\frac{\epsilon_t/\epsilon_{t0}}{0.31 (f'_t)^2 (\epsilon_t/\epsilon_{t0})^{1.7} + \epsilon_t/\epsilon_{t0}} \right] & \epsilon_{t0} \leq \epsilon_t \end{cases} \quad (25)$$

where f_t and ϵ_t are the tensile stress and strain of concrete, respectively, and f'_t is the peak tensile stress at strain ϵ_{t0} .

In addition, the parameters for plasticity are required to define the plastic flow behavior and yield function [60]. Those parameters include the dilation angle (ψ), the flow potential eccentricity (e), the ratio of the equibiaxial to uniaxial compressive stress at the point of initial yield (f_{b0}/f_{c0}), the ratio of the second stress invariant on the tensile meridian to that on the compressive meridian at initial yield (K_c), and the viscosity parameter (μ). The values of the plasticity model parameters used in this study are provided in Table 5. It is worth noting that the modulus of elasticity of concrete (E_c) is computed as $E_c = 4700(f'_c)^{0.5}$ according to ACI Code 318-19 [63].

The combined isotropic-kinematic hardening constitutive relationship is employed to model the cyclic behavior of steel. The evolution law of this model consists of the isotropic and kinematic components. The isotropic hardening, represented by the change in the yield surface size, is described as:

$$\sigma^0 = \sigma|_0 + Q_\infty \left(1 - e^{-b\bar{\epsilon}^{pl}} \right) \quad (26)$$

where $\sigma|_0$ is the initial yield stress at plastic strain $\bar{\epsilon}^{pl} = 0$, Q_∞ is the peak size of the yield surface that can achieve, and b is the change

rate of the size of the yield surface with plastic strain. The kinematic component represents the translation of the yield surface in stress space through the backstress α , the change rate of which ($\dot{\alpha}$) is given as:

$$\dot{\alpha}_k = C_k \frac{1}{\sigma_0} (\sigma - \alpha) \dot{\bar{\epsilon}}^{pl} - \gamma_k \alpha_k \dot{\bar{\epsilon}}^{pl} \quad (27)$$

where C_k and γ_k are the initial kinematic hardening modulus and the decreasing rate of the kinematic hardening modulus with increasing plastic strain, respectively, and k is the number of the backstress. The total backstress is computed as:

$$\alpha = \sum_{k=1}^N \alpha_k \quad (28)$$

In this study, three backstress components and one backstress are used to describe the kinematic hardening rule of Q235 steel [41] and HRB 400 steel reinforcement [64], respectively. The model parameters for the steels used in this paper are listed in Table 6. As the Rigid pads and PT tendons behaved elastically during the tests, both components are simply represented using elastic material model.

5.2. Validation of the FE model

Fig. 21(a) compares the hysteresis curves of Auxetic-Peanut-1 between experiment and FE modeling. The numerical results exhibit a strong correlation with the experimental data, thereby confirming the efficacy of the developed numerical model developed in this study. However, the initial stiffness predicted by the FE modeling is slightly larger than the experimental result. This discrepancy can be attributed to the idealized “tie” constraints established between the T-shaped supports and the RC foundation in the model. The pier-foundation connection is considered as a fully welded connection before connection decompression, while the connection may have some flexibility in the tests. Furthermore, the residual displacement upon unloading computed by the FE modeling is smaller than the test results. This disparity arises from the fact that the numerical model underestimates the additional lateral load resistance from ASSPDs during unloading. In addition, the FE model is also incapable of considering the PT loss that caused by PT anchorage seating, thus underrates the restoring force for pier re-centering. It is noteworthy that the predicted positive and negative maximum loads at 3% drift are 58.0 kN and 59.1 kN, respectively. These values correspond to relative differences of 5.7% and 4.2%, respectively, when compared to the experimental values.

Fig. 21(b) compares the variation in PT force between the test and numerical simulation of Auxetic-Peanut-1. The behavior of the PT tendons is shown to be predominantly elastic, with only minimal PT force loss. The introduction of ECC and the reinforced steel plates in Segment 3 contributes to reduced damage and slight prestress loss. At maximum drift, the numerical PT force is 1.33 times the initial PT force, while the experimental PT force is 1.24 times the initial value. Fig. 21(c) shows the strain history of longitudinal steel reinforcements at Segment 3 of Auxetic-Peanut-1. The numerical model effectively captures the strain response of the steel reinforcing rebars at various locations. In general, the finite element model developed in this study exhibits reasonable accuracy in predicting the local and component response of the specimen and can be deemed reliable for further analysis.

5.3. Parametric study

This section aims to investigate the effects of key components in a SC-PSBP, specifically focusing on the connecting steel plate and the PT tendons with various initial PT forces. Additionally, the geometry of

Table 6
Material model parameters of steels used in the finite element models.

Steel	σ_{l0}/MPa	Q_{∞}/MPa	b	C_1	γ_1	C_2	γ_2	C_3	γ_3
HPB400 (8 mm diameter)	534	267	0.1	7500	500	–	–	–	–
HPB400 (12 mm diameter)	468	234	0.1	7500	500	–	–	–	–
HPB400 (20 mm diameter)	467	233	0.1	7500	500	–	–	–	–
Q235	226	10	1.2	100 000	1250	7500	50	500	0

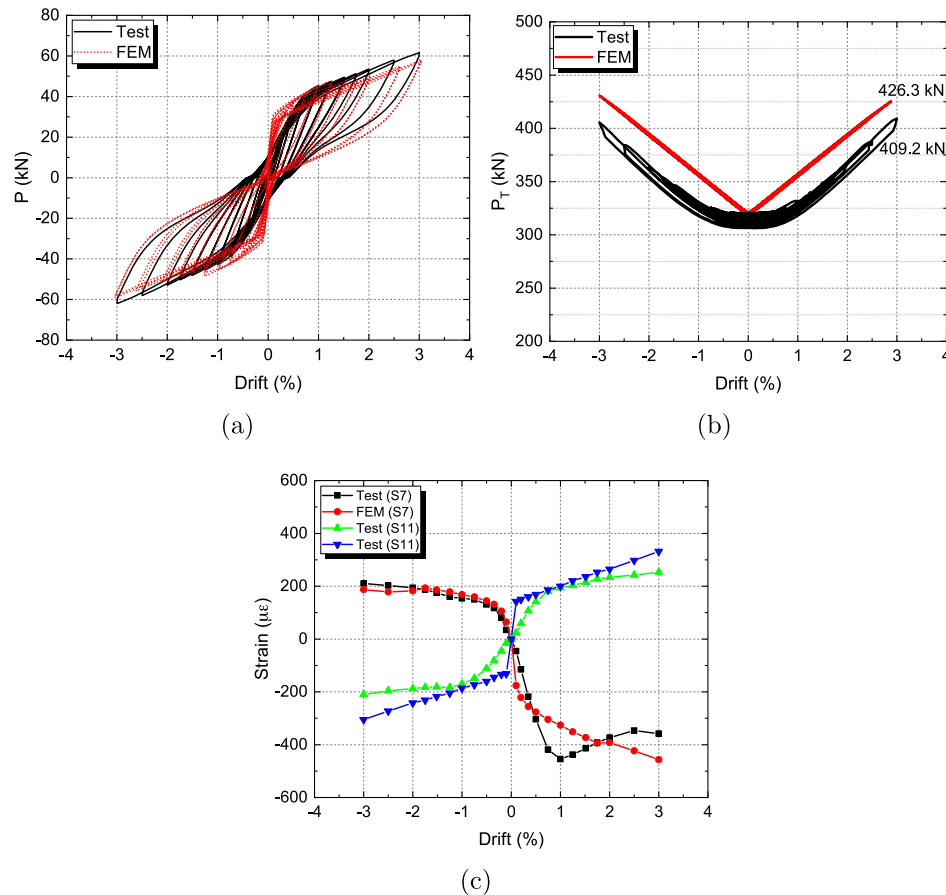


Fig. 21. Comparison of hysteresis response of Auxetic-Peanut-1 between experiments and numerical modeling, (a) hysteresis curves, (b) PT force vs. drift, and (c) strain in the longitudinal steel reinforcement.

ASSPDs significantly influences the seismic performance of SC-PSBPs. Parametric studies have been conducted to quantitatively assess the impact of design parameters related to ASSPDs on the cyclic response of SC-PSBPs. The design geometric parameters of ASSPD considered in this study include r/R , t/R , r_1/r_2 , and r_2/D , while the initial PT force remains constant at 320 kN. The design parameters for peanut-shaped and ellipse-shaped ASSPDs are presented in Table 7.

5.3.1. Effects of ASSPDs and connecting steel plates

To determine the influence of connecting steel plates (CSPs) on the hysteresis response of SC-PSBPs, three models were developed, i.e., one without CSPs, one without ASSPDs, and the reference model, Auxetic-Peanut-1. Fig. 22(a) clearly demonstrates the significant impact of the CSPs between Segment 1 and 2, and 2 and 3 on the energy dissipation and self-centering capacity of the SC-PSBP. The model without CSPs shows minimal energy dissipation, with a gap opening at the Segment 2–3 connection instead of at the base of the pier (Fig. 23(a)). This can be attributed to the lower decompression moment at the intermediate segment connections (pure PT connections) as compared to that at the base, and the applied moment firstly surpasses the decompression

moment at Segment 2–3 connection. This rocking mechanism prevents significant plastic straining of the ASSPDs, thus reducing the energy dissipation capacity of the SC-PSBP. Interestingly, the strength of the model without CSPs is comparable to that of the reference model, and even exhibits slightly higher load-bearing capacity at a maximum drift of 3.0%.

The hysteresis behavior of the model without ASSPDs is similar to that of the model without CSPs (Fig. 22(a)). Both models display a typical bilinear elastic behavior with minimal energy dissipation. As the gap opens at the pier base (Fig. 23(b)), the lateral stiffness of the model without ASSPDs is significantly reduced and is lower than the other two models. Among the three models, this model exhibits the smallest post-decompression strength. It is evident that ASSPDs contribute more significantly to the overall strength of SC-PSBPs compared to CSPs. While CSPs have a minimal impact on the strength, stiffness, and energy dissipation of SC-PSBPs, they are still essential for the novel self-centering precast pier. The placement of CSPs facilitates the anticipated rocking mechanism in SC-PSBPs, specifically the rocking at the base adjacent to the ASSPDs.

Table 7

Geometry of the elliptical and peanut-shaped perforated unit cells used in the FE modeling.

Model ^a	Geometric dimension/mm						Nondimensional parameter				
	r	R	t	r_1	r_2	D	ϕ	r/R	t/R	r_1/r_2	r_2/D
OE-0.6-0.06	30.7	51.2	3.1	–	–	–	0.64	0.60	0.06	–	–
OE-0.6-0.12	28.7	47.8	5.7	–	–	–	0.56	0.60	0.12	–	–
OE-0.6-0.18	26.9	44.9	8.1	–	–	–	0.49	0.60	0.18	–	–
OE-0.6-0.34	23.3	38.8	13.0	–	–	–	0.37	0.60	0.34	–	–
OE-0.6-0.30	24.0	40.0	12.0	–	–	–	0.39	0.60	0.30	–	–
OE-0.2-0.18	11.3	56.4	10.2	–	–	–	0.26	0.20	0.18	–	–
OE-0.4-0.18	20.0	50.0	9.0	–	–	–	0.41	0.40	0.18	–	–
OE-0.8-0.18	32.6	40.7	7.3	–	–	–	0.54	0.80	0.18	–	–
OE-1.0-0.18	37.3	37.3	6.7	–	–	–	0.56	1.00	0.18	–	–
OP-1.1-0.39	–	–	12.5	12.1	11.0	30.8	0.45	–	–	1.10	0.39
OP-1.0-0.55	–	–	11.6	12.1	12.1	21.9	0.48	–	–	1.00	0.55
OP-0.9-0.39	–	–	7.5	12.1	13.2	30.8	0.62	–	–	0.90	0.39
OP-0.85-0.39	–	–	4.9	12.1	14.3	30.9	0.71	–	–	0.85	0.39
OP-0.8-0.39	–	–	2.6	12.1	15.4	30.7	0.80	–	–	0.80	0.39
OP-1.0-0.39	–	–	10.1	12.1	12.1	30.7	0.53	–	–	1.00	0.39
OP-1.0-0.31	–	–	10.3	12.1	12.1	39.6	0.52	–	–	1.00	0.31
OP-1.0-0.46	–	–	10.6	12.1	12.1	26.4	0.51	–	–	1.00	0.46
OP-1.0-0.34	–	–	9.9	12.1	12.1	35.1	0.54	–	–	1.00	0.34

^a The models OE-0.4-0.18, OE-0.8-0.18, OP-1.0-0.39, and OP-1.0-0.31 have the same design parameters as the test specimen Auxetic-Ellipse-2, Auxetic-Ellipse-1, Auxetic-Peanut-1, and Auxetic-Peanut-2, respectively.

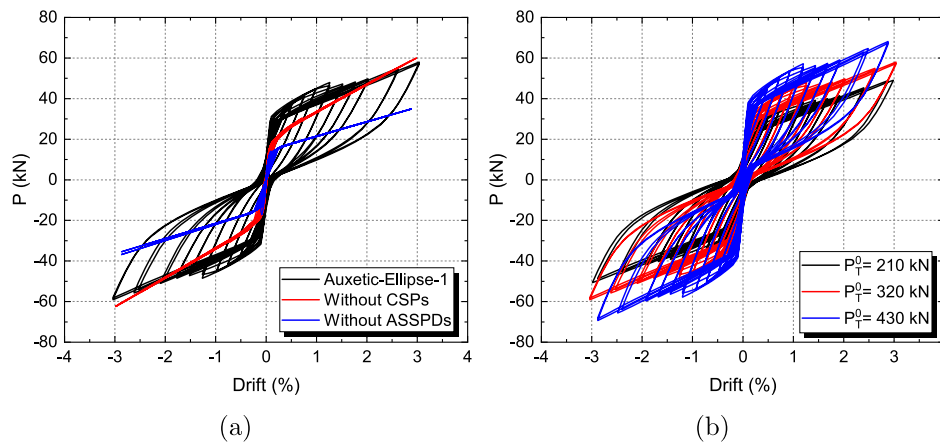


Fig. 22. Effects of steel components on hysteresis response of Auxetic-Peanut-1, (a) effects of connecting steel plates (CSPs) and ASSPDs, and (b) effect of initial PT force.

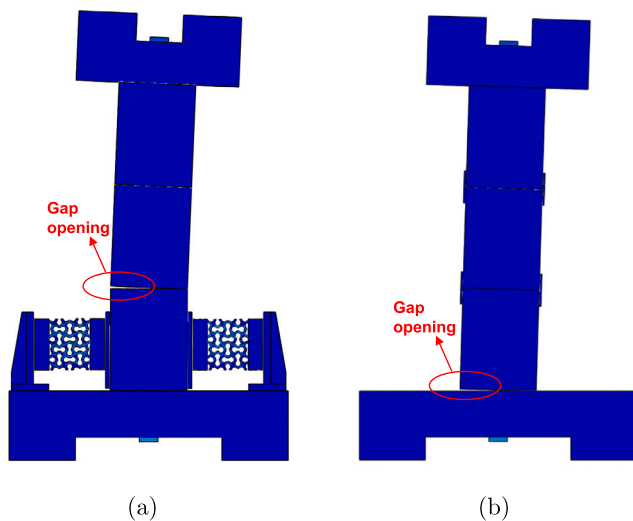


Fig. 23. Deformed state of Auxetic-Peanut-1, (a) without connecting steel plates (CSPs), and (b) without ASSPDs.

The effect of the thickness of CSPs on SC-PSBPs is studied using three numerical models. The three models are identical to Auxetic-Ellipse-1 except the CSPs that have thicknesses of 3 mm, 6 mm, and 10 mm, respectively. Fig. 24(a) shows the hysteresis response of the three models. Remarkably, the three hysteresis curves are almost identical, indicating that a change in the thickness of the CSP upon a minimum magnitude has a negligible impact on the overall behavior of SC-PSBPs. Furthermore, Fig. 24(b) illustrates the local stress response of the CSPs, which displays significant variations among the three models. These differences are attributed to the opening and sliding of joint gaps in different models and between different segment joints in a given model. Nevertheless, it is important to note that the amount of gap opening at each intermediate segment joint is minimal, and all CSPs exhibit elastic behavior. Thus, even the thinnest CSP (3 mm in thickness) used in the numerical model provides sufficient restraints on segment joint opening and sliding.

5.3.2. Effect of initial PT force

Three different values of ($P_T^0 = 210, 320, 430$ kN) are selected to study the effect of initial PT force on the hysteresis response of SC-PSBP, which corresponding to different axial compression ratios ($n = 0.12, 0.18, 0.24$), respectively. Fig. 22(b) illustrates the hysteresis curves of SC-PSBPs applied with different initial PT forces. All the models present the flag-shaped hysteresis response. As the initial PT force increases, the load bearing capacity improves and the residual drift

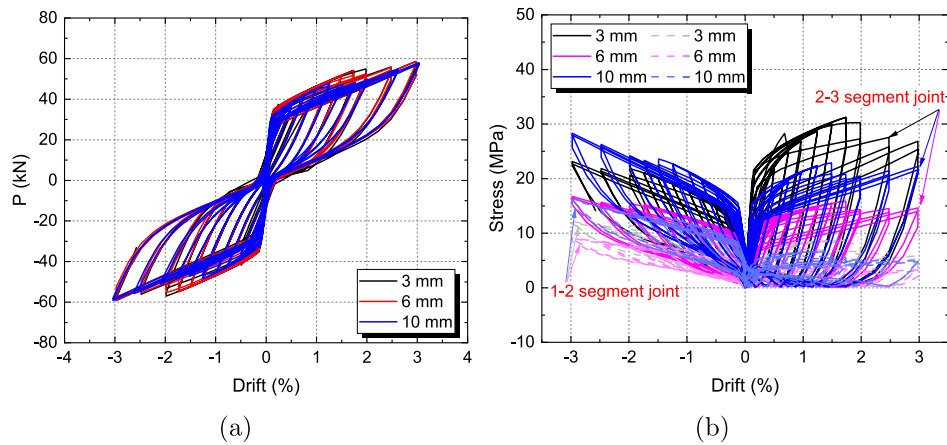


Fig. 24. Effects of thickness of connecting steel plate on the hysteresis response of SC-PSBPs, (a) influence on the hysteresis response, and (b) stress development at the CSPs with various thickness.

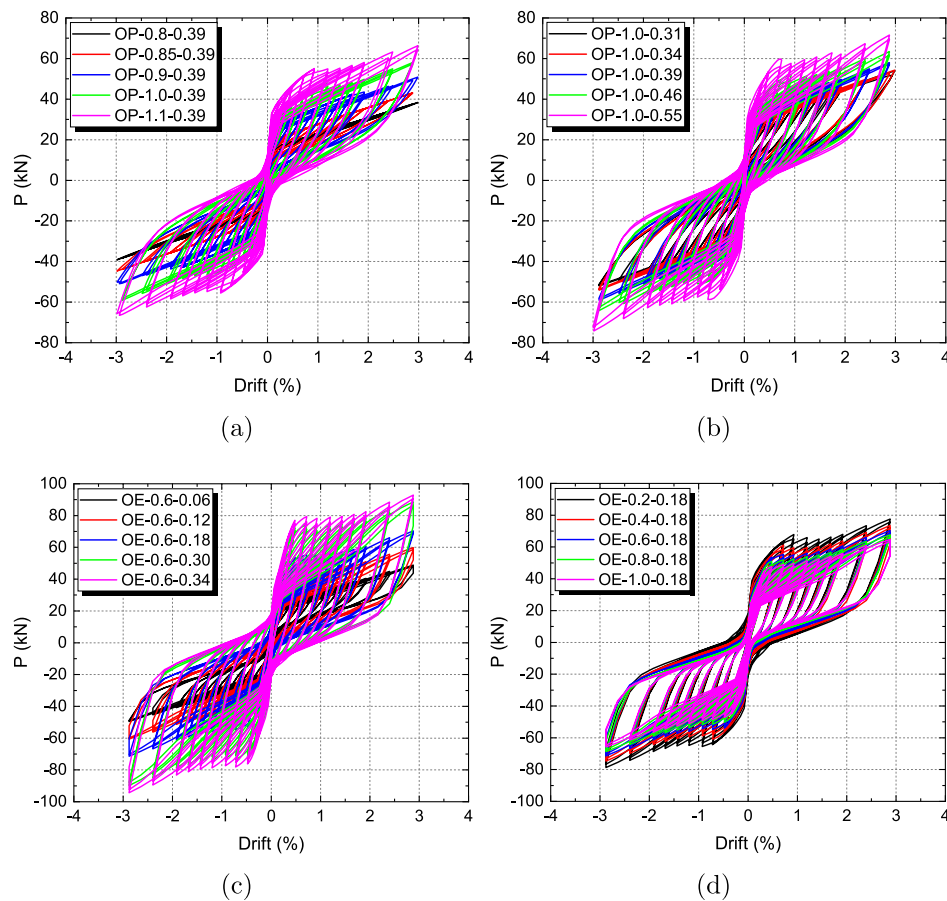


Fig. 25. Hysteresis curves of the models with various ASSPD perforation parameters, (a) with various r_2/r_1 , (b) with various r_2/D , (c) with various t/R , and (d) with various r/R .

decreases. For the model with $P_T^0 = 430$ kN, the residual displacement upon unloading at a 3% drift circle decreases by 5 mm as compared to the model with $P_T^0 = 210$ kN.

5.3.3. Effects of geometric parameters of ASSPD infill plate

Effect of aspect ratio of peanut-shaped hole r_2/r_1 . Fig. 25(a) illustrates the hysteresis response of SC-PSBPs equipped with ASSPDs that have the same r_2/D value of 0.39 but different r_2/r_1 values of 1.1, 1, 0.9, 0.85, and 0.8. It is apparent that as r_2/r_1 decreases, the porosity of the infill plate in ASSPD increases significantly. The model OP-0.8-0.39 had the infill plate porosity with a nearly 78% increase as compared

to OP-1.1-0.39. This increase in porosity is accompanied by a 57% reduction in the load bearing capacity at maximum drift and a decrease in energy dissipation capacity to 1/25 of that observed in OP-1.1-0.39 (see Fig. 26(a)).

Effect of the ratio of arc radius to peanut-shaped hole major axis r_2/D . Fig. 25(b) depicts the hysteresis curves of the SC-PSBPs equipped with various ASSPDs of the same r_2/r_1 value of 1.0 and varying r_2/D values of 0.31, 0.34, 0.39, 0.46, and 0.55. The alteration in r_2/D exhibits minimal influence on the porosity but results in a significant change in the minimum thickness m between adjacent unit cells. In comparison

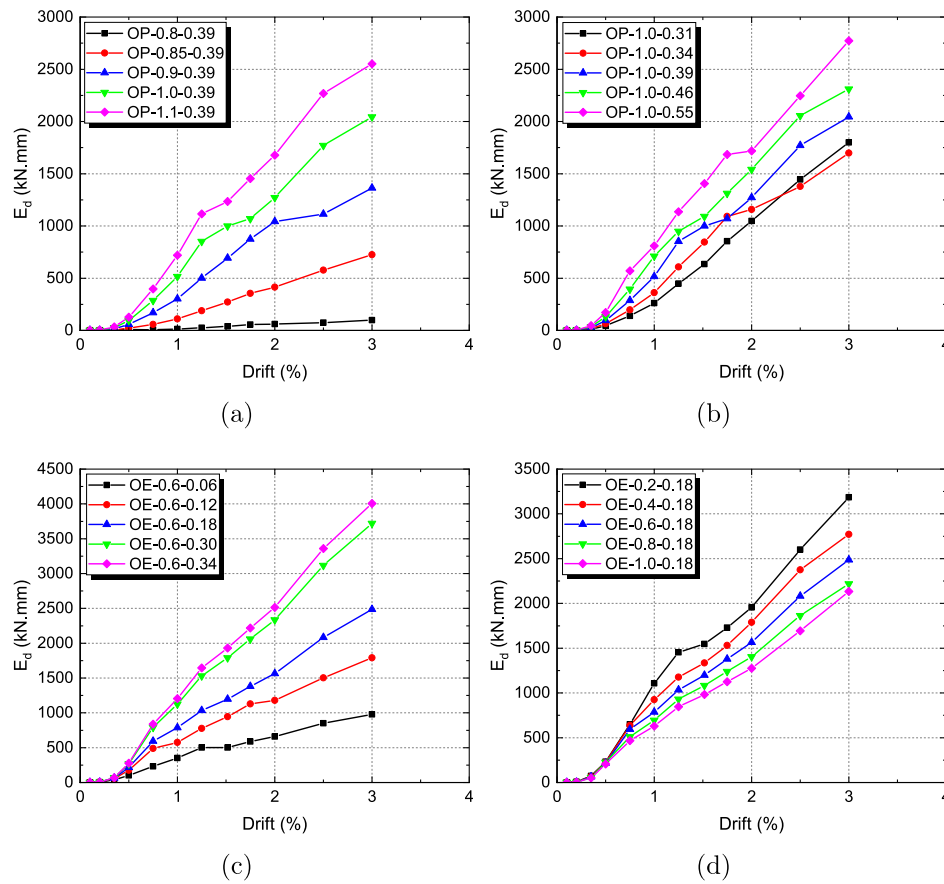


Fig. 26. Energy dissipation of the models with various ASSPD perforation parameters, (a) with various r_2/r_1 , (b) with various r_2/D , (c) with various t/R , and (d) with various r/R .

to model OP-1.0-0.31, OP-1.0-0.55 demonstrates a remarkable increase of 32% in bearing capacity and 54% in energy dissipation at maximum drift (Figs. 25(b) and 26(b)). This suggests that by selecting an optimal value of r_2/D while maintaining a constant infill plate porosity, the strength and energy dissipation capacity of ASSPDs can be effectively optimized. Consequently, the novel SC-PSBPs incorporating peanut-hole ASSPDs offer tunability in their hysteresis behavior through the optimal design of the auxetic perforated unit cells.

Effect of the ratio of ligament thickness to elliptical hole length t/R . The models OE-0.6-0.06, OE-0.6-0.12, OE-0.6-0.18, OE-0.6-0.3, and OE-0.6-0.34 have the same r/R value and various t/R values of ASSPD infill plates. An increase in t/R corresponds to a decrease in the infill plate porosity 7. The content of the solid matrix in the infill plate of an ASSPD increases with the thickness of the ligament between neighboring perforations, resulting in an enhancement in the bearing capacity and energy dissipation capacity (Figs. 25(c) and 26(c)). The strength and dissipated energy of OE-0.6-0.34 at the maximum loading drift were 92.9 kN and 4006 kN mm larger than those of OE-0.6-0.12, respectively. On the other hand, the residual deformation of the model at a given drift reduces with t/R , which is beneficial to the self-centering capacity of SC-PSBPs.

Effect of aspect ratio of elliptical hole r/R . The effect of the parameter r/R was investigated by comparing the hysteresis response of the models OE-0.2-0.18, OE-0.4-0.18, OE-0.8-0.18, and OE-1.0-0.18, which had ASSPDs with the same t/R value of 0.18 and varying r/R values of 0.2, 0.4, 0.8, and 1.0. It is worth mentioning that a change in r/R significantly influences the porosity of the infill plate of ASSPD. An increase in r/R gives rise to a degradation of strength and energy dissipation capacity (Figs. 25(d) and 26(d)). The change in the porosity

of the infill plate may be a crucial influence that leads to different hysteresis responses in those SC-PSBP specimens.

6. Conclusions

This paper introduces a novel self-centering precast segmental bridge piers (SC-PSBPs) that utilize innovative replaceable auxetic steel shear panel dampers (ASSPDs). Each precast segment is connected through post-tensioning (PT) tendons, ASSPDs, and connecting steel plates. The seismic performance of SC-PSBPs is investigated through an experimental program and numerical modeling. Two groups of SC-PSBP specimens, one with elliptical perforated ASSPDs and the other with peanut-shaped perforated ASSPDs, undergo quasi-static cyclic loading tests. Additionally, 3D finite element (FE) models are developed to deepen the understanding of the effects of various design parameters. The main conclusions of this study are as follows:

- The precast segments likely behave without any damage, while the infill plates of the auxetic steel shear panel dampers exhibit significant in-plane and out-of-plane permanent deformation throughout the testing. Both groups of SC-PSBPs exhibit a stable hysteresis response. The addition of out-of-plane buckling restrainers to the ASSPD infill plates enhances the strength and energy dissipation capacity, albeit with larger residual displacements. The largest residual drifts are about 10% and 28% for Auxetic-Ellipse and Auxetic-peanut series of specimens, respectively, at the 3% drift cycle.
- The specimens demonstrate a desired rocking mechanism, with the primary gap forming at the pier-foundation connection. The plastic deformation of ASSPD infill plates increases with the gap

angle, providing the main energy dissipation capacity of SC-PSBPs. Furthermore, the total gap angles at the intermediate segment connections are less than 0.3%, negligible compared to that at the base connection (about 2%) at the 3% drift cycle. The connecting steel plates between neighboring precast segments increase the connection decompression moment.

- Auxetic-Ellipse-1 and Auxetic-Peanut-1, with similar porosity but different perforation shapes, exhibit varying strength (54.1 kN and >61.6 kN, respectively) and deformation capacities. At 2% drift, fractures predominantly occur at the ligaments between neighboring perforations in Auxetic-Ellipse-1, resulting in reduced ductility, energy dissipation, and equivalent viscous damping compared to Auxetic-Peanut-1. It is likely that ASSPDs with peanut-shaped perforations offer SC-PSBPs a more stable hysteresis performance.
- The numerical model without connecting steel plates exhibits a different rocking mechanism from the tests, with gap openings observed at the intermediate segment joint and negligible plastic deformation in ASSPDs, underscoring the significance of connecting steel plates. An increasing in the CSP thickness upon 3 mm has little impact on the hysteresis response of SC-PSBP. Increasing the initial PT force from 210 kN to 430 kN enhances the bearing capacity by 40% while reducing residual displacement by 100% at the 3% drift cycle. Furthermore, an increase in the relative thickness of the ligament between neighboring holes from 0.06 to 0.34 and a decrease in the hole aspect ratio in ASSPDs from 1.1 to 0.85 significantly improve the strength of SC-PSBPs by 90% and 57%, respectively.
- This paper primarily focuses on the single mechanism of the proposed SC-PSBPs, which includes rocking only at the bottom joint. However, exploring the potential of allowing rocking and rocking-sliding at multiple intermediate segment joints as energy dissipation options could lead to great seismic performance. This area can be further researched in future studies. Moreover, other types of metamaterial-based energy dissipation devices, different from those used in this paper, can be employed to extend the application of novel materials and structures to self-centering precast bridge piers.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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